# Mapping the Storm

AP Human Geography · Unit 1 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

**No stimulus.** You're interning at a Gulf Coast county's emergency-management office before hurricane season. Your supervisor wants you to master the geographer's toolkit: maps, locations, regions, and how people and the environment shape each other.

**A. Identify the type of region that "the South" in the United States is an example of.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names it: a perceptual (vernacular) region
- Describes it: exists in people's minds, with fuzzy borders

MODEL ANSWER

"The South" is a perceptual (vernacular) region: it exists in people's minds, so its borders are fuzzy and change depending on who you ask.

**Common mistake:** "The South is a formal region because it is made up of states." States have official borders, but "the South" doesn't: no law says which states are in it. That makes it perceptual, not formal.

**Why the model works:** It names the exact type (perceptual, or vernacular) and shows the borders are fuzzy and live in people's minds.

**B. Describe how the Mercator projection distorts the size of Greenland compared with the size of Africa.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the distortion: Greenland looks about as big as Africa
- Gives the truth or the reason: Africa is about 14 times bigger, or Mercator stretches areas near the poles

MODEL ANSWER

**T1-** The Mercator projection stretches the area of places near the poles, so they look much bigger than they really are.
**E2-** On a Mercator map, Greenland looks about as big as Africa, but Greenland is about 2.2 million km² while Africa is about 30 million km², roughly 14 times larger.
**A3-** Because Greenland is near the North Pole and Africa straddles the equator, people who use only Mercator maps can badly underestimate Africa's true size.

**Common mistake:** "Greenland looks way bigger on the Mercator map." It only says what the map SHOWS. Add the truth (Africa is actually about 14 times bigger) and why (Mercator stretches places near the poles).

**Why the model works:** It describes the distortion, gives real numbers (2.2 vs. 30 million km²), and explains it with latitude.

**C. Describe the difference between a city's site and its situation, using New Orleans or another real city as your example.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes SITE: the physical features of the place itself
- Describes SITUATION: location relative to other places and its connections

MODEL ANSWER

**T1-** A city's site is the physical land it sits on, while its situation is its location compared to other places.
**E2-** New Orleans's site is low, swampy land in a bend of the Mississippi River, and about half the city is below sea level. Its situation is near the river's mouth on the Gulf of Mexico, which makes it a port linking the central United States to world trade.
**A3-** The site explains New Orleans's flood risk, while the situation explains why people built a major city there anyway.

**Common mistake:** "Site is the city's exact address, and situation is what's going on in the city, like the economy." Two mix-ups: an exact address is absolute location, and situation isn't "what's happening." Site is the physical land; situation is its connections.

**Why the model works:** It defines both terms, gives proof for each (swampy land vs. a river-to-Gulf port), and shows why the difference matters.

**D. Explain how a county government could use remote sensing to respond to a hurricane after it makes landfall.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says what remote sensing collects: images from satellites, aircraft, or drones
- Explains the response: finding flooding or damage, then sending rescue, supplies, or repairs there

MODEL ANSWER

**T1-** A county could use remote sensing, which collects images of Earth from satellites, planes, or drones, to see storm damage without sending people into danger.
**E2-** After a hurricane, before-and-after images show flooded neighborhoods, destroyed homes, and blocked roads; NOAA, for example, flies planes over the coast to photograph hurricane damage.
**A3-** Knowing where the worst damage is lets the county send rescue crews, water, and generators there first, saving time and lives.

**Common mistake:** "Remote sensing would help the government know about the hurricane." Too vague. Say what remote sensing collects (images from above) and what the county would do with them.

**Why the model works:** It says what remote sensing is, what it shows after a storm, and how that changes the response (help goes to the worst areas first).

**E. Explain ONE reason a functional region, such as a hospital's service area or a city's commuter zone, can change in size over time.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: new transportation, a competing node, or new technology
- Explains how the edge moves: people come from farther (it grows) or switch to a closer node (it shrinks)

MODEL ANSWER

**T1-** A functional region can grow when new transportation makes its node faster to reach.
**E2-** For example, when Washington, D.C.'s Metro Silver Line was extended to Dulles Airport and Loudoun County, Virginia, in 2022, people in those outer suburbs could ride a train into the city.
**A3-** Because a functional region is defined by connections to its node, faster connections let people come from farther away, so the commuter zone's edge moves outward and the region grows.

**Common mistake:** "A functional region changes because the government redraws its borders." That's how a formal region, like a county, changes. A functional region has no official border: it changes when connections to its node change.

**Why the model works:** It names a cause (a new rail line), gives a real example (the Silver Line, 2022), and explains the effect (the region grows).

**F. Explain ONE difference between environmental determinism and possibilism, using a real-world example.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes determinism: the environment controls or limits what people and societies can do
- Describes possibilism with an example: people choose how to adapt, often with technology

MODEL ANSWER

**T1-** Environmental determinism says the environment controls how people live and develop, while possibilism says it only sets limits and people choose how to respond.
**E2-** For example, about a quarter of the Netherlands lies below sea level, but the Dutch built dikes, pumps, and polders (land drained from the sea) to farm and live there.
**A3-** A determinist would expect such flood-prone land to hold a society back, but the Netherlands became one of the world's wealthiest countries, which supports possibilism.

**Common mistake:** "Environmental determinism is when the environment determines things, and possibilism is when anything is possible." It just repeats the terms with no example, and possibilism doesn't say ANYTHING is possible: nature still sets limits.

**Why the model works:** It defines both ideas, gives a real example (the Netherlands), and explains how it supports possibilism.

**G. Explain ONE limitation of using only quantitative data, such as census numbers, to understand a place after a hurricane, and how qualitative data could help.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a limitation: numbers don't show people's reasons, feelings, or experiences
- Explains what qualitative data adds: interviews, field observation, or stories

**MODEL ANSWER**

**T1-** One limitation of quantitative data is that numbers show how many people were affected, but not why they made their choices.
**E2-** For example, the 2010 census counted about 140,000 fewer people in New Orleans than in 2000, after Hurricane Katrina, but it can't show why some families returned and others never did.
**A3-** Qualitative data, like interviews with residents and field observation, could reveal reasons such as rebuilding costs or ties to family and church, helping leaders plan a better recovery.

**Common mistake:** “Quantitative data is numbers and qualitative data is words.” Correct definitions, but no answer. Explain what numbers MISS after a hurricane, and how interviews or field observation fill that gap.

**Why the model works:** It names what numbers miss (the why), gives a real example (New Orleans after Katrina), and explains what interviews would add.

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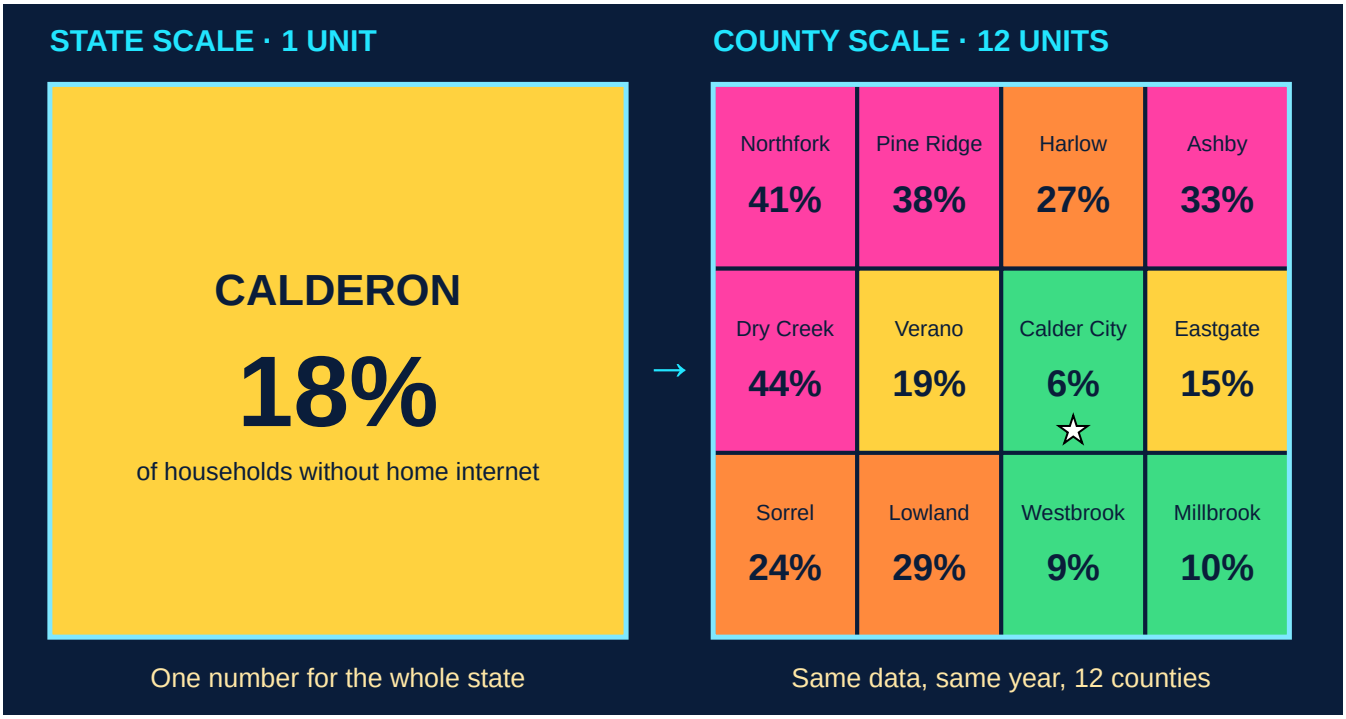
Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# Two Scales, One State

AP Human Geography · Unit 1 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Both maps show the same variable for the same fictional state in the same year. The left panel reports one value for the whole state. The right panel divides Calderon into its 12 counties. The capital, Calder City, is marked with a star.

**Percent of Households Without Home Internet, State of Calderon (fictional), 2020: State Scale vs. County Scale**




0–10% 11–20% 21–30% 31–45% ★ State capital (Calder City)

*Data and places are invented for practice. The state figure is a population-weighted average: most of Calderon's people live in the metro counties around Calder City, so those low rates pull the state number down. Shapes simplified, not to scale.*

**A. Identify the type of thematic map used in the county-scale panel of the stimulus.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names it: a choropleth map
- Describes the method: areas shaded or colored by data class

MODEL ANSWER

It is a choropleth map: each county is shaded one color based on which percent class it falls in.

**Common mistake:** "It is a political map of Calderon's counties." A political map shows borders and names, not data. This map colors each county by a percent, which makes it a thematic map with a specific name.

**Why the model works:** It names the map type precisely and adds a quick phrase showing it knows how the map works.

**B. Using the map, describe how the pattern of households without home internet changes between the state scale and the county scale.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the state scale as one value or one shade (18%) for the whole state
- Describes the county-scale pattern with evidence: a wide range (6% to 44%), low near the capital or metro, high in rural or edge counties

MODEL ANSWER

**T1-** At the state scale there is no pattern to see, because the whole state is one shade with a single value of 18%.
**E2-** At the county scale the same data range from 6% in Calder City and 9% in Westbrook, clustered around the capital, up to 41% in Northfork and 44% in Dry Creek along the western and northern edges.
**A3-** Zooming in reveals a clustered pattern of rural counties where more than a third of households lack internet, which the single state number completely hides.

**Common mistake:** "At the county scale there are more colors and more details." That's about the picture, not the pattern. A describe answer needs the actual values (18% vs. 6% to 44%) and WHERE the high and low counties are.

**Why the model works:** It describes the state scale (one value, 18%), gives specific county evidence with places and numbers, and names what the zoom-in reveals (a rural cluster the state figure hides).

**C. Explain ONE reason rural counties in Calderon have higher rates of households without home internet than the counties around the capital.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a reason: low population density / homes spread far apart, high cost of infrastructure per customer, distance from the city, or lower incomes
- Explains the cause-and-effect chain to the higher rate

**MODEL ANSWER**

**T1-** Rural counties have higher rates of households without internet because their low population density makes internet infrastructure very expensive per customer.
**E2-** In Dry Creek (44%) homes are spread miles apart, so a company must lay far more fiber-optic cable to reach each household than it does in dense Calder City (6%), where one line serves hundreds of homes.
**A3-** Because the company earns less for every mile of cable in a rural county, it builds there last or not at all, so many rural households are left without service and the rate stays high.

**Common mistake:** “Rural counties don't have internet because they are in the country.” That restates the question: 'rural' and 'in the country' mean the same thing. You need the reason (homes far apart, costly cable) and the chain that leads to fewer connections.

**Why the model works:** It names the reason (low density, high cost per customer), proves it with map numbers (Dry Creek 44% vs. Calder City 6%), and explains the chain that leaves rural homes unconnected.

**D. Using the map, explain ONE limitation of using only the state-level data to make decisions about improving internet access in Calderon.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- States the limitation: one state value is an average that hides or masks differences between counties
- Uses map evidence and explains the effect on a decision (wrong places, spread too thin, underestimating need)

**MODEL ANSWER**

**T1-** One limitation is that a single state-level number is an average that hides the differences inside the state, so planners can't see where the problem actually is.
**E2-** The map shows Calderon at 18%, but at the county scale Dry Creek is 44% and Northfork is 41% while Calder City is only 6%, and eight of the 12 counties are above the state figure.
**A3-** Because the 18% figure makes the whole state look moderately connected, a government using it alone might spread money evenly or underestimate the need, instead of targeting counties like Dry Creek and Northfork, where more than 40% of households lack internet.

**Common mistake:** “The state data is a limitation because it is from 2020 and might be out of date.” The question asks about using STATE-LEVEL data, so the limitation should be about scale, not the year. Compare 18% to the county values and explain what gets hidden.

**Why the model works:** It names what's hidden (county differences), proves it with map numbers (18% vs. 44% and 6%), and explains the decision that could go wrong.

**E. Explain how a state government agency could use a geographic information system (GIS) to decide where to build new broadband internet infrastructure.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Describes what GIS does: layers, overlays, or combines different spatial data sets to analyze them
- Explains how that analysis leads to a location decision (where need is highest, where building is cheapest, which places to prioritize)

**MODEL ANSWER**

**T1-** A state agency could use GIS to layer several kinds of spatial data on top of each other and find the best places to build.

**E2-** For example, it could overlay this county map of households without internet with layers showing existing fiber-optic lines, population density, household income, and the locations of schools and clinics.

**A3-** Where the layers overlap, such as a rural county like Northfork with 41% unconnected, no nearby fiber, and a school, the agency can see where need is highest and building is possible, so it can target its limited money there first.

**Common mistake:** "The government could use GPS to find out where people live and then build internet there." That's GPS (finding locations), not GIS (layering and analyzing data). It also skips how the agency would decide WHICH places get built first.

**Why the model works:** It says what GIS does (layers data), names specific layers tied to the map, and explains how the overlap leads to choosing where to build.

**F. Describe ONE way a dot density map or a graduated symbol map of the same data would show the pattern differently from the county choropleth.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names the map type: dot density or graduated/proportional symbol
- Describes a real difference: it shows counts (number of households) instead of a rate, so metro counties could look bigger; or dots show where people cluster inside a county

**MODEL ANSWER**

**T1-** A dot density map would show the number of households without internet rather than the percent in each county.
**E2-** If each dot stood for 500 households and the dots were placed where people live, Calder City might have many dots even though its rate is only 6%, while Dry Creek at 44% might have only a few dots because so few people live there.
**A3-** This matters because it shows where the largest total number of unconnected households is, which can be a different place from where the rate is highest, and the dots can hint at clustering inside a county instead of coloring the whole county one shade.

**Common mistake:** "A dot density map would use dots instead of colors, so it would look different." That describes the drawing style, not the pattern. Say what the dots would SHOW: counts instead of percents, and where the dots would pile up.

**Why the model works:** It names the map type, describes exactly what would change (counts vs. rates, with Calder City and Dry Creek as proof), and says why that matters.

**G. Explain how improved internet access in Calderon's rural counties is an example of time-space compression, and explain ONE effect it could have on a rural community.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Connects to time-space compression: distance matters less because places are connected faster or instantly
- Explains one effect on the rural community (remote jobs, telehealth, online school, online businesses, people staying rather than leaving)

**MODEL ANSWER**

**T1-** Better internet is an example of time-space compression because it shrinks the time it takes to connect a rural place with distant ones, so distance matters less.
**E2-** A family in Dry Creek that once drove two hours to see a specialist in Calder City could have a video visit in minutes, and a student there could take an online course from a university anywhere.
**A3-** One effect is that rural residents can reach jobs, health care, and education without leaving, which could help young people stay in the community and let local businesses sell to customers far away, slowing rural population loss.

**Common mistake:** “Time-space compression means the world is getting smaller because of technology.” That's a definition, not an application. You need to connect it to internet in Calderon's rural counties AND explain one effect on the community.

**Why the model works:** It ties internet access to the concept (shrinking the time to connect across distance), gives concrete examples from Dry Creek, and explains a community effect with a cause-and-effect chain.

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Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# Fewer Babies, Bigger Cities

AP Human Geography · Units 2 · 6 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

The table shows data for four countries. (The countries are fictional and the data were created for practice.)

**Selected Population and Development Indicators**

| Country | Total fertility rate 1995 | Total fertility rate 2020 | Female secondary school enrollment, 2020 (%) | Urban population, 2020 (%) | Infant mortality, 2020 (per 1,000 births) |
|---|---|---|---|---|---|
| **Norvale** | 6.1 | 5.4 | 22 | 24 | 68 |
| **Kestria** | 5.8 | 3.1 | 61 | 48 | 31 |
| **Tamboa** | 3.4 | 1.9 | 88 | 71 | 12 |
| **Ruzhan** | 2.0 | 1.3 | 97 | 82 | 3 |

*Total fertility rate = average number of children per woman. Data created for practice.*

**A. Identify the country in the table that is most likely in Stage 2 of the demographic transition model.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Norvale

MODEL ANSWER

Norvale.

**Common mistake:** "Ruzhan." Ruzhan has the LOWEST TFR (1.3). That fits Stage 4 or 5, not Stage 2.

**Why the model works:** Norvale has the highest TFR (5.4) and infant mortality and is the most rural: all Stage 2 signs.

**B. Describe the relationship between female secondary school enrollment and total fertility rate shown in the table.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- States an inverse/negative relationship (more enrollment, lower TFR)
- Supports it with numbers from the table

MODEL ANSWER

**T1-** The table shows an inverse relationship: as female secondary enrollment rises, TFR falls.
**E2-** Norvale has only 22% enrollment and a TFR of 5.4, while Ruzhan has 97% enrollment and a TFR of just 1.3.
**A3-** This matters because it suggests that keeping girls in school is one of the strongest ways a country lowers its birth rate.

**Common mistake:** "Education and fertility are related." It doesn't say HOW they're related and uses no numbers from the table.

**Why the model works:** It names the direction (inverse) and backs it up with data from two countries.

**C. Explain ONE reason why increased access to education for women leads to lower fertility rates.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a mechanism: delayed marriage, paid work, family-planning knowledge, or decision-making power
- Explains how that mechanism results in fewer children

MODEL ANSWER

**T1-** Women who stay in school longer tend to have fewer children because they start families later and have more career options.
**E2-** A girl who finishes secondary school usually marries and has her first child years later, and she is more likely to work for wages.
**A3-** Because she starts later and each child costs her more in lost income (opportunity cost), she has fewer children overall.

**Common mistake:** "Educated women are smarter, so they have fewer kids." Not a real mechanism. It doesn't explain WHAT changes in her life to lower fertility.

**Why the model works:** It picks one mechanism (marrying later + working) and walks through how it leads to fewer births.

**D. Using data from the table, explain how urbanization can contribute to declining fertility rates.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses table data showing more-urban countries have lower TFR
- Explains why: children cost more or aren't farm labor in cities, or cities offer better access to health care/contraception

MODEL ANSWER

**T1-** More-urban countries have lower fertility because children cost more and help less in cities.
**E2-** In the table, Norvale is 24% urban with a TFR of 5.4, while Tamboa is 71% urban with a TFR of 1.9.
**A3-** In cities, children don't provide farm labor and housing and schooling cost more, so children become an economic cost rather than an asset, which leads families to have fewer.

**Common mistake:** “Cities are crowded, so people have fewer kids.” No data from the table, and 'crowded' isn't the real reason. Explain costs, jobs, or health care.

**Why the model works:** It quotes the table AND explains the asset-to-cost shift.

**E. Describe ONE likely change in the shape of Kestria's population pyramid resulting from the change in its total fertility rate between 1995 and 2020.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes a shape change: narrower base, or a wider working-age middle
- Links it to Kestria's falling TFR (5.8 → 3.1)

MODEL ANSWER

**T1-** Kestria's pyramid will get a narrower base because fewer children are being born.
**E2-** Its TFR fell from 5.8 in 1995 to 3.1 in 2020, so each new age group at the bottom is smaller than the one before.
**A3-** Meanwhile the large group born in the 1990s moves up into working ages, so the pyramid becomes a slimmer triangle with a bulge moving up into the working ages. It is not yet the beehive shape of a country near replacement level, because a TFR of 3.1 is still well above 2.1.

**Common mistake:** “Kestria's pyramid will change shape.” It says the shape changes but not HOW. Say which part gets narrower or wider.

**Why the model works:** It links the data (5.8 → 3.1) to a specific shape change (narrower base, wider middle).

**F. Explain ONE economic challenge Ruzhan is likely to face as a result of its total fertility rate.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Recognizes that TFR is below replacement (≈2.1), so the population ages or shrinks
- Explains an economic challenge: labor shortage, pension/health costs, smaller tax base, dependency ratio

MODEL ANSWER

**T1-** Ruzhan will face an aging population with too few workers because its fertility is below replacement level.
**E2-** Its TFR of 1.3 is well below 2.1, so fewer young people enter the workforce each year while more people retire.
**A3-** This leads to labor shortages and a smaller tax base to pay for pensions and health care, raising the elderly dependency ratio.

**Common mistake:** "Ruzhan's population will go down." That's a population change, not an economic challenge, and it doesn't explain the effect on jobs or money.

**Why the model works:** It connects low TFR → aging → a specific economic problem (labor shortages, pension costs).

**G. Explain ONE limitation of using national-scale data, such as the data in the table, to understand fertility patterns.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names variation hidden by national averages (urban/rural, regional, income, ethnic/religious)
- Explains why that hides important patterns or leads to wrong conclusions

MODEL ANSWER

**T1-** One limitation is that national data are averages that hide differences within a country.
**E2-** For example, Kestria's TFR of 3.1 could combine cities near 2.0 with rural regions above 5.0.
**A3-** Because of this, the table can't show where fertility is actually high or why, so policies based on it could miss the places that need them most.

**Common mistake:** "The table doesn't have enough countries." That's about sample size, not scale. The question asks what national-level data hides.

**Why the model works:** It names the hidden variation (cities vs. rural areas) and explains why that leads to wrong conclusions.

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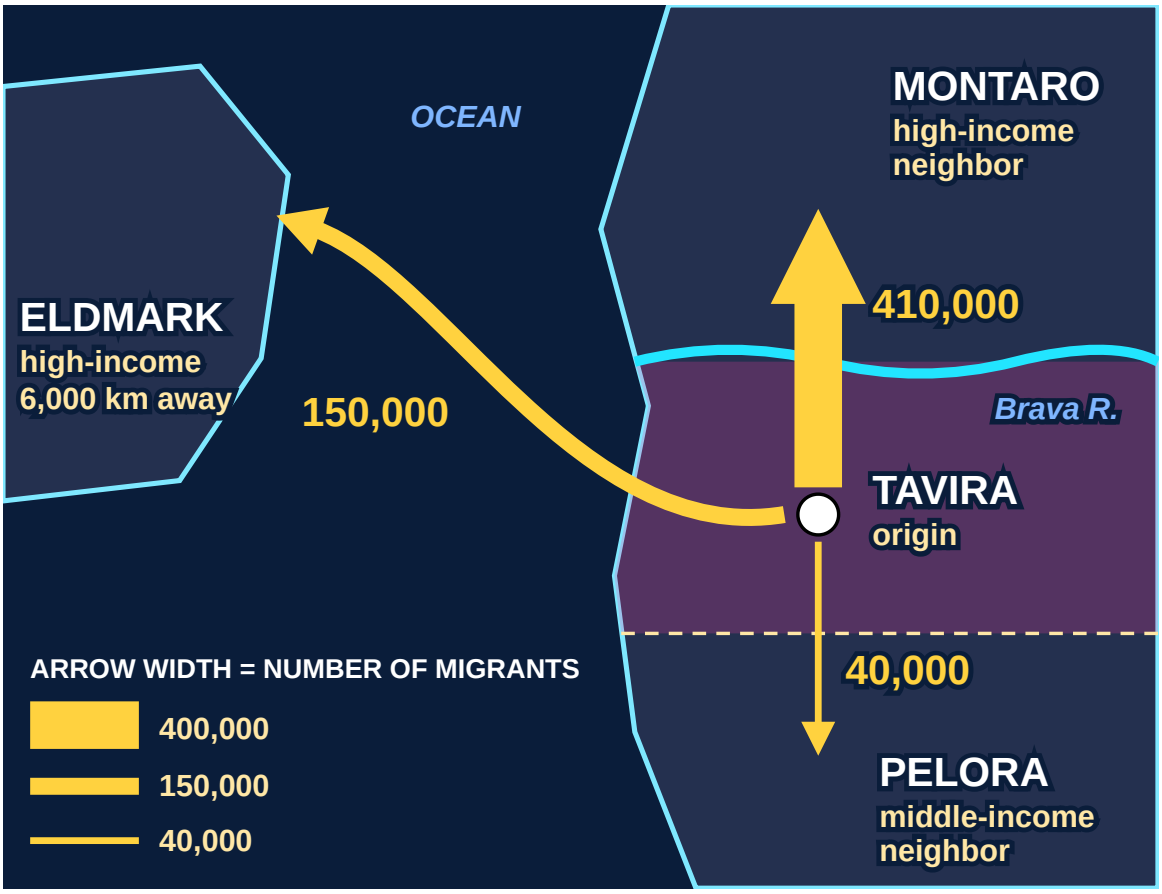
# On the Move

AP Human Geography · Unit 2 · Two stimuli (like Question 3) · 7 points · Teacher key · GeoArcade FRQ Tutor

Stimulus 1 is a flow map of the people who left the country of Tavira in 2023. Stimulus 2 is one of those migrants describing her move in her own words. (Tavira, its neighbors, the numbers, and the person are fictional and were created for practice.)

**Stimulus 1: Emigration from Tavira, 2023 (flow map) · Stimulus 2: One Migrant's Story**

**Stimulus 1: Emigration from Tavira, 2023**




*Flow map created for practice. Tavira, Montaro, Eldmark, Pelora, and all numbers are fictional. Tavira's population is about 20 million. Schematic, not to scale.*

**Stimulus 2: "Why I Left Tavira"**

"The drought lasted three years. Our maize died, and the family farm could not feed us anymore. My cousin Elena had already lived in Montaro for six years. She found me a job at the fruit-packing plant where she works. Getting here was the hardest part: we crossed the Brava River at night, where people drown every year, and then walked for hours along the border wall to find a way through. Now I send money home every month so my little brothers can stay in school. Someday, when the rains return, I will go back to Tavira."

*Account by Marisol, age 24, who moved from Tavira to Montaro in 2023. Marisol and her words are fictional and were created for practice.*

**A. Identify ONE environmental push factor described in Stimulus 2.** IDENTIFY

**EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:**

- Drought / lack of rain
- Crop failure on the family farm

**MODEL ANSWER**

A drought that lasted three years and killed the maize on Marisol's family farm.

**Common mistake:** "She wanted to make more money." That's economic, and it's really a pull toward Montaro. The question asks for an ENVIRONMENTAL push: something in nature that drove her away.

**Why the model works:** It names an environmental push (the drought) and ties it to passage details (three years, dead maize).

**B. Using Stimulus 1, describe the pattern of migration from Tavira and how it reflects the role of distance in migration.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Describes the pattern: most went a short distance to neighboring Montaro
- Uses map evidence: numbers or arrow widths (410,000 vs. 150,000)
- Connects to distance: distance decay or Ravenstein

**MODEL ANSWER**

**T1-** Most migrants from Tavira moved a short distance to a neighboring country, and far fewer made the long trip overseas.
**E2-** In 2023, 410,000 people went next door to Montaro, while only 150,000 crossed the ocean to Eldmark, about 6,000 km away, even though both are high-income countries.
**A3-** This fits Ravenstein's law that most migrants move short distances, an example of distance decay: long moves cost more and feel riskier, so fewer people make them.

**Common mistake:** "Most people went to Montaro." It names the pattern but stops: no numbers and no link to distance. Add 410,000 vs. 150,000 and explain why distance matters.

**Why the model works:** It describes the pattern (most went next door), proves it with map numbers, and links it to Ravenstein and distance decay.

**C. Using Stimulus 2, explain how Marisol's move is an example of chain migration.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Shows what chain migration is: following family or people from home who already moved
- Uses the passage: cousin Elena already lived there and found her a job
- Explains how the link helped cause the move

MODEL ANSWER

**T1-** Marisol's move is chain migration because she followed a family member who had already settled in Montaro.
**E2-** Her cousin Elena had lived there for six years and found Marisol a job at the fruit-packing plant where she works.
**A3-** Because a relative and a job were waiting, the move was less risky and Montaro was the obvious choice, and now Marisol can help the next relative follow, so the chain keeps growing.

**Common mistake:** "Marisol moved to Montaro to get a job." True, but it doesn't show the CHAIN: no mention that her cousin was already there or found her that job.

**Why the model works:** It defines chain migration, uses the passage (cousin Elena, the job), and explains why that link made the move easier.

**D. Explain how ONE intervening obstacle described in Stimulus 2 affects migration from Tavira to Montaro.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names an obstacle from the passage: the Brava River or the border wall
- Explains its effect (dangerous, costly, fewer people go)

MODEL ANSWER

**T1-** One intervening obstacle is the Brava River, a physical barrier between Tavira and Montaro.
**E2-** Marisol says her group crossed it at night, and people drown there every year.
**A3-** Because the crossing is so dangerous, the river discourages some Tavirans from leaving at all and makes the trip deadly for others, so it slows and reduces migration even though Montaro is right next door.

**Common mistake:** "The border is between Tavira and Montaro." It points to a location but explains nothing. Name the obstacle (the river or the wall) and explain how it makes migration harder.

**Why the model works:** It names an obstacle from the passage (the Brava River), uses her details (night crossing, drownings), and explains how it slows and reduces migration.

**E. Using BOTH stimuli, explain ONE economic effect of remittances on Tavira.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses Stimulus 2: Marisol sends money home every month
- Uses Stimulus 1: many migrants work in high-income countries
- Explains an economic effect on Tavira (income, spending, schooling, businesses, dependence)

MODEL ANSWER

**T1-** Remittances, the money migrants send home, raise household incomes in Tavira and bring money into its economy.
**E2-** Marisol sends money home every month so her brothers can stay in school, and Stimulus 1 shows 410,000 Tavirans moved to Montaro, a high-income country with higher wages.
**A3-** When hundreds of thousands of migrants send money like Marisol does, families can afford food, school, and housing, and their spending supports local businesses, though Tavira can also become dependent on money earned abroad.

**Common mistake:** "Marisol sends money home to her family." Only half the evidence, and no effect. Add the map's numbers and explain what the money does to Tavira's economy.

**Why the model works:** It uses BOTH stimuli (Marisol's monthly money, the 410,000 in high-income Montaro) and explains the effect on Tavira's incomes and spending.

**F. Explain ONE way large-scale emigration, like the migration shown in Stimulus 1, can change the population structure of the origin country.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says who leaves: young or working-age adults (or mostly men)
- Explains the change in structure (pinched pyramid, higher dependency ratio, fewer births, sex imbalance)

MODEL ANSWER

**T1-** Large-scale emigration can leave the origin with fewer working-age adults and a higher dependency ratio, because most migrants are young adults.
**E2-** In 2023 alone, 600,000 people left Tavira, and migrants like 24-year-old Marisol are in their prime working and child-bearing years, so the young-adult bars of Tavira's population pyramid would shrink.
**A3-** With fewer workers to support children and the elderly, the dependency ratio rises, and with fewer young adults left to have children, the birth rate can fall too.

**Common mistake:** "Tavira's population will go down." True, but that's SIZE, not structure. Say WHO leaves (young, working-age adults) and how that changes the mix left behind.

**Why the model works:** It says who leaves (young adults), uses evidence (600,000 people, Marisol at 24), and explains the effect (higher dependency ratio, fewer births).

**G. Explain ONE limitation of using Stimulus 1 to understand why people leave Tavira.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- States a limitation: the map shows numbers and places, not reasons
- Explains why it matters, with an example (drought, wages, violence; you'd have to guess)

**MODEL ANSWER**

**T1-** One limitation is that the flow map shows how many people left Tavira and where they went, but not WHY.
**E2-** It shows 410,000 people moving to Montaro, but not whether they were pushed by drought, low wages, or violence, while Stimulus 2 reveals that Marisol left because of a drought and followed her cousin.
**A3-** Because the map has no information about push and pull factors, someone using it alone would have to guess the causes and could get them wrong.

**Common mistake:** “The map doesn't show every country in the world.” That's about what's on the map, not about why people leave. Explain that the map shows numbers and places but no push or pull factors.

**Why the model works:** It names the limitation (no reasons), gives examples the map can't show, and explains why it matters (you'd have to guess).

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Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# English Goes Global

AP Human Geography · Unit 3 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

The English language is spoken in many countries, but at very different rates. India and China each have more than 1.3 billion people. India has more English speakers than China, but the number of English speakers is growing in both countries.

**Number of English Speakers in Selected Countries, 2018 (population above age five)**

| Country | Number of English speakers | Percent of population who speak English |
|---|---|---|
| **China** | 10 million | less than 1% |
| **India** | 125 million | 12% |
| **Netherlands** | 15 million | 90% |
| **Nigeria** | 79 million | 53% |
| **Philippines** | 92 million | 95% |
| **Sweden** | 8 million | 86% |
| **United Kingdom** | 60 million | 98% |
| **United States** | 296 million | 96% |

*Source: Ethnologue. Remember the class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).*

**A. Describe ONE historical reason for India's current number of English speakers.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a historical reason: British colonial rule of India
- Describes how English spread or stayed: used in government, schools, or business, and kept after independence

MODEL ANSWER

**T1-** India has many English speakers today because it was a British colony.
**E2-** Under the British Raj (1858–1947), English was the language of government, courts, and universities, so educated Indians had to learn it.
**A3-** After independence, India kept English as an official language, partly as a neutral link between its many regional languages, so it is still widely spoken.

**Common mistake:** “India speaks English because it is a big country.” Size doesn't explain language. China is just as big and has under 1% English speakers. You need the history.

**Why the model works:** It names the history (British colonial rule), gives specific proof (government, courts, universities), and explains why English lasted after 1947.

**B. Explain ONE contemporary reason why people in China and India are learning English.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a present-day reason: globalization, business, jobs, studying abroad, the internet, or tourism
- Explains how English helps (more opportunities, higher pay, communicating with the world)

MODEL ANSWER

**T1-** People in China and India are learning English because it is the global lingua franca of business.
**E2-** Multinational companies, tech firms, and call centers hire workers who can speak English with customers and partners around the world.
**A3-** Because English skills open these better-paying jobs, learning English can raise a person's income and help the country compete in the global economy.

**Common mistake:** "English is the best language." That's an opinion, not a reason, and it doesn't explain what English does for people in China or India.

**Why the model works:** It names a present-day reason (global business), gives an example (jobs with multinational companies), and explains the payoff (higher income).

**C. Describe ONE barrier to the diffusion of English throughout China.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a barrier: government policy, Mandarin emphasis, internet limits, language difference, few teachers, rural access, no British colonial history, or cultural pride
- Describes how it slows English from spreading in China

MODEL ANSWER

**T1-** One barrier to English spreading in China is government control over information.
**E2-** China's "Great Firewall" blocks many English-language websites, and the government promotes Mandarin as the national language.
**A3-** Because people in China have fewer chances to see and use English online or in daily life, English spreads slowly, which helps explain why fewer than 1% of people there speak it.

**Common mistake:** "English is hard." Too vague. Hard for whom, and why? Name a specific barrier in China and describe how it slows English.

**Why the model works:** It names a specific barrier (government control), gives proof (the Great Firewall, Mandarin policy), and describes the effect (fewer chances to use English).

**D. Using the table, explain ONE reason for the difference in the percentage of English speakers in the Netherlands compared with the percentage in China.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses the table: the Netherlands is 90% vs. China under 1%
- Explains a reason: similar language, trade/globalization, closeness to the UK, media, schools, or China's huge Mandarin-speaking market

MODEL ANSWER

**T1-** The Netherlands has a much higher share of English speakers (90%) than China (less than 1%) partly because Dutch and English are closely related languages.
**E2-** Both are Germanic languages in the Indo-European family, so they share many words and grammar patterns, while Mandarin uses tones and characters.
**A3-** Because English is much easier for Dutch speakers to learn, and they hear it constantly through trade and English-language TV, a far larger percentage of Dutch people become fluent.

**Common mistake:** "The Netherlands has more English speakers than China." That just restates the table. It gives no reason at all. The question asks you to explain WHY the percentages are so different.

**Why the model works:** It quotes the table, names a reason (language similarity), and explains how it leads to more English speakers.

**E. Explain ONE economic reason why a corporation based in an English-speaking country would locate a call center in a country where English is not the dominant language.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names an economic reason: lower wages/labor costs, a large English-speaking workforce, time zones, or tax incentives
- Explains how it benefits the corporation (saves money, raises profits, 24-hour service)

MODEL ANSWER

**T1-** A corporation might put a call center in a country like India because labor costs are much lower there.
**E2-** India has about 125 million English speakers, so companies can hire many workers who speak English but earn far lower wages than workers in the United States.
**A3-** Paying lower wages for the same service cuts the company's costs, which raises its profits.

**Common mistake:** "Because people in other countries need jobs." That's about the workers, not the corporation. The question asks for an economic reason that benefits the company.

**Why the model works:** It names the economic reason (lower labor costs), uses the table (125 million English speakers in India), and explains the payoff (higher profits).

**F. Explain ONE limitation of using the data in the table to understand a country's level of economic development.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- States a limitation: language data doesn't measure income, GDP per capita, HDI, health, or education (or it's one year / national averages)
- Explains why that's misleading, ideally with an example (the Philippines is 95% but less developed than China)

MODEL ANSWER

**T1-** One limitation is that the table measures how many people speak English, not how wealthy or developed a country is.
**E2-** For example, the Philippines is 95% English-speaking, but it has a much lower GDP per capita than China, where less than 1% speak English.
**A3-** Because English ability doesn't track development, using this table alone could lead someone to the wrong conclusion about which countries are more developed.

**Common mistake:** "The table doesn't show every country." That's about how many countries are listed, not about measuring development. Explain what kind of data is missing.

**Why the model works:** It names what's missing (economic data), gives a clear example (the Philippines vs. China), and explains how it could mislead.

**G. Explain ONE negative local impact of a global lingua franca on places where a different language is spoken.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a negative impact: language loss or extinction, loss of local culture, cultural homogenization, or inequality for non-English speakers
- Explains how the lingua franca causes it locally

MODEL ANSWER

**T1-** One negative local impact of English as a global lingua franca is the loss of local languages.
**E2-** When young people use English for school, jobs, and the internet, they may stop speaking their family's language, so fewer children learn it.
**A3-** As a local language fades, the stories, traditions, and knowledge tied to it can be lost, weakening the place's cultural identity.

**Common mistake:** "English is bad for other countries." Too vague and too broad. Name a specific local impact and explain how the lingua franca causes it.

**Why the model works:** It names a specific impact (language loss), explains how it happens (young people switch to English), and why it matters (traditions and identity).

# Maple Street District

AP Human Geography · Unit 3 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

The passage below describes how a single neighborhood in a mid-sized U.S. city changed over one hundred years as different groups moved in and out. Read it closely before you answer.

**A Century on Maple Street: One Neighborhood, Three Communities**

**1920s.** In the 1920s, Maple Street District was the German heart of Ridgeway, a mid-sized Midwestern city. Immigrants from Saxony and Hanover filled its brick row houses. They built St. Johannes Lutheran Church on the corner of Maple and Third, opened beer gardens along the avenue, and printed a weekly German-language newspaper, the *Ridgeway Zeitung*.

**1970s.** By the 1970s, most German families had moved to the suburbs. Mexican and Central American families arrived, drawn by factory jobs. The old church building reopened as Our Lady of Guadalupe Catholic Parish. Taquerías and a panadería replaced the beer gardens, a Spanish-language radio station went on the air, and residents began an annual Día de los Muertos festival on Maple Street.

**2020s.** In the 2020s, young professionals moved in. Coffee shops and a yoga studio opened, and rents rose sharply. The festival is now sponsored by the city and draws thousands of tourists, though several long-running restaurants have closed.

*Maple Street District and the city of Ridgeway are fictional and were created for practice.*

**A. Identify the type of diffusion that brought German culture to Maple Street District in the 1920s.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Relocation diffusion
- Spread through migration: people moved and brought their culture with them

MODEL ANSWER

Relocation diffusion: German immigrants moved to Ridgeway and brought their language, religion, and food with them.

**Common mistake:** “Contagious diffusion, because it spread from person to person.” Contagious diffusion is when an idea spreads between neighbors who stay put. Here the Germans themselves moved to Ridgeway, so the culture traveled WITH people. That's relocation diffusion.

**Why the model works:** It names the exact type (relocation) and shows the evidence: immigrants moved and carried their culture with them.

**B. Using the passage, describe ONE way the cultural landscape of Maple Street District changed between the 1920s and the 1970s.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a specific landscape feature from the passage (church, beer gardens, taquerías, newspaper, radio, signs, foods)
- Describes the change: what it was before and what it became

**MODEL ANSWER**

**T1-** The cultural landscape changed as the German community was replaced by a Mexican and Central American one, an example of sequent occupance.
**E2-** The passage says St. Johannes Lutheran Church reopened as Our Lady of Guadalupe Catholic Parish, and taquerías and a panadería replaced the beer gardens along the avenue.
**A3-** The buildings stayed on the same corners, but their names, foods, and religion changed to match the new residents, so each group left its own visible layer on Maple Street.

**Common mistake:** "The neighborhood got more Mexican people." That's a change in WHO lived there, not in the landscape. The question wants something you could see on the street, like the church becoming a Catholic parish or taquerías replacing beer gardens.

**Why the model works:** It names the overall change (sequent occupance), uses specific before-and-after details from the passage (the church, the beer gardens), and explains what the layers show.

**C. Explain how the Día de los Muertos festival in the 2020s is an example of a cultural practice changing as it spreads to new groups of people.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names how the festival changed: city-sponsored, tourists, commercial/commodified, mainstream, bigger, or adopted by outsiders (acculturation or stimulus diffusion)
- Explains why spreading to new groups changed its meaning or purpose (cause → effect)

**MODEL ANSWER**

**T1-** The Día de los Muertos festival shows how a cultural practice can change as it spreads to new groups, a process called stimulus diffusion, along with commodification.
**E2-** The passage says residents began the festival for their own neighborhood in the 1970s, but by the 2020s it is sponsored by the city and draws thousands of tourists.
**A3-** As outsiders adopt the celebration, its purpose shifts from honoring family ancestors to entertaining visitors and bringing money into the city, so the festival keeps its name but loses some of its original religious meaning.

**Common mistake:** "The festival is more popular now." That says it changed, but not HOW it spread to new people or WHY that changed its purpose. Add who runs it now (the city, for tourists) and what that does to the meaning.

**Why the model works:** It names the process (acculturation/commodification), quotes the shift from residents to city sponsorship and tourists, and explains how a new audience changes the festival's purpose.

**D. Using the passage, describe ONE way religion is visible in the cultural landscape of Maple Street District.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a religious feature from the passage: the church/parish (Lutheran, Catholic, Guadalupe) or the Día de los Muertos festival
- Describes how it is visible or what it shows about the residents (building, corner, name, celebrated in the street, which faith)

**MODEL ANSWER**

**T1-** Religion is visible on Maple Street through its place of worship.
**E2-** The passage says German immigrants built St. Johannes Lutheran Church at the corner of Maple and Third, and the same building later reopened as Our Lady of Guadalupe Catholic Parish.
**A3-** A church building and its name tell anyone walking by which faith the residents practice, so the landscape shows the shift from German Protestants to Mexican and Central American Catholics.

**Common mistake:** "The people in the neighborhood are religious." That tells us about the people, not the landscape. The question wants something visible: name the church (or the festival) and describe what it shows.

**Why the model works:** It names the feature (a church), gives the exact detail from the passage (St. Johannes → Our Lady of Guadalupe), and describes what the building reveals about the residents.

**E. Explain ONE centrifugal effect that the changes of the 2020s could have on long-time residents of Maple Street District.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a dividing effect on long-time residents: displacement, being pushed or priced out, losing gathering places, feeling like outsiders, tension or resentment with newcomers, loss of identity (gentrification)
- Explains the cause → effect using the 2020s changes (rising rents, new businesses, closures, newcomers, tourists)

**MODEL ANSWER**

**T1-** The changes of the 2020s could act as a centrifugal force by making long-time residents feel their community is being taken from them.
**E2-** The passage says rents rose sharply, coffee shops and a yoga studio opened, and several long-running restaurants closed as young professionals moved in.
**A3-** Because the new businesses serve newcomers and tourists rather than the families who built the neighborhood, older residents may feel like outsiders on their own street, which creates tension between the two groups and divides the community, a pattern geographers connect to gentrification.

**Common mistake:** “Rents went up, so it costs more to live there.” That's the economic fact, but it isn't a centrifugal effect yet. Finish the chain: higher rents close family restaurants and push residents out, so the community feels divided and loses its identity.

**Why the model works:** It names the centrifugal effect (feeling pushed out, tension), quotes the 2020s changes from the passage, and explains how those changes divide old-timers from newcomers.

**F. Explain how the Spanish-language radio station is an example of a centripetal force for the community that arrived in the 1970s.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names the unifying effect: unites or connects people, builds shared identity or belonging, preserves language and culture, strengthens community
- Explains how the station does it: Spanish language, news, music, information, events, listening together

**MODEL ANSWER**

**T1-** The Spanish-language radio station is a centripetal force because it unites the Mexican and Central American community around a shared language and identity.
**E2-** The station broadcasts news, music, and announcements in Spanish, so residents from different countries all hear about neighborhood events like the Día de los Muertos festival and local businesses in their own language.
**A3-** Because people share the same information and culture every day, the station strengthens their sense of belonging and keeps the community connected, even as the neighborhood changes around them.

**Common mistake:** “The radio station plays Spanish music.” That's what it does, not what it does FOR the community. Add the effect: hearing shared news and music in Spanish unites residents and strengthens their identity.

**Why the model works:** It names the unifying effect (shared language and identity), gives concrete examples of what the station broadcasts, and explains how that builds belonging and keeps the community connected.

**G. Explain ONE way globalization or popular culture could threaten the local culture of Maple Street District, AND ONE way technology could help residents preserve it.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a threat from globalization or popular culture: chains or global brands, tourists, commercialization, homogenization, local businesses replaced or lost
- Names a way technology could preserve local culture: internet, social media, streaming, recording, online archive, apps, videos, podcasts

**MODEL ANSWER**

**T1-** Globalization and popular culture can threaten Maple Street's local culture, but technology can also help residents protect it.

**E2-** The passage shows coffee shops and a yoga studio opening for newcomers and tourists while long-running family restaurants close; if global chains keep arriving, the district could start to look like any other trendy neighborhood, a step toward cultural homogenization.

**A3-** At the same time, residents could stream the Spanish-language radio station online and post festival videos and family recipes on social media, so young people and families who moved away can still take part, keeping the traditions alive even if the storefronts change.

**Common mistake:** "Globalization makes everything the same, so the neighborhood will lose its culture." Good start, but that's only half the question. You also need ONE way technology could help preserve the culture, like streaming the radio station or posting festival videos online.

**Why the model works:** It names the threat (chains and tourists replacing local businesses → homogenization), names a technology fix (streaming the station, posting videos and recipes), and explains the effect of each.

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Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

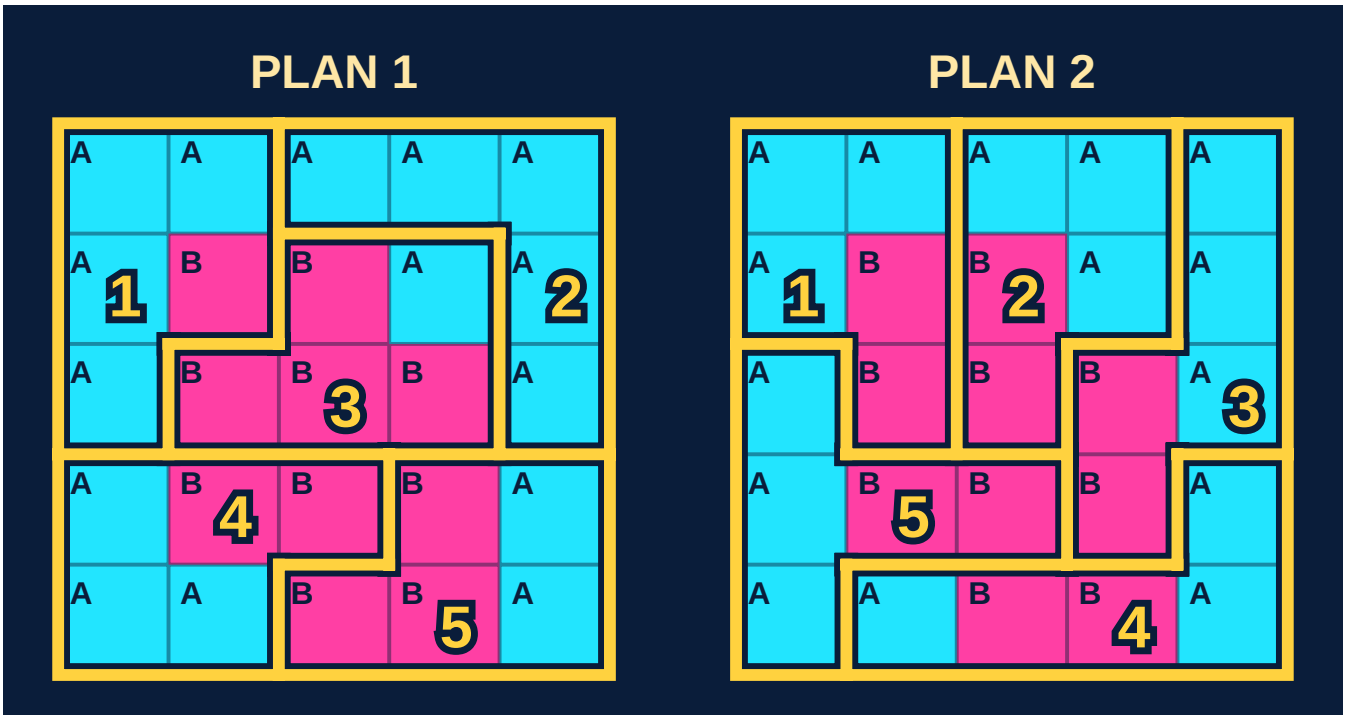
# Drawing the Lines

AP Human Geography · Unit 4 · Two stimuli (like Question 3) · 7 points · Teacher key · GeoArcade FRQ Tutor

Westmoor, a fictional U.S. state, has 25 voting precincts with equal populations and 5 seats in the U.S. House of Representatives. After the census, mapmakers proposed two plans for grouping those precincts into 5 districts, using the same voters and votes.

**Westmoor Redistricting: Two District Plans (Stimulus 1) and Election Results (Stimulus 2)**

**Stimulus 1: Two District Plans for Westmoor**




A = precinct won by Party A  B = precinct won by Party B  District border  1–5 = district number

*Fictional state created for practice. Simplified: all 25 precincts have equal populations, and a party wins a district by winning at least 3 of its 5 precincts.*

**Stimulus 2: Election Results in Westmoor Under Each Plan**

| Result | Party A | Party B |
|---|---|---|
| **Statewide vote share** | 60% | 40% |
| **Precincts won (of 25)** | 15 | 10 |
| **Districts won under Plan 1 (of 5)** | 3 | 2 |
| **Percent of seats under Plan 1** | 60% | 40% |
| **Districts won under Plan 2 (of 5)** | 5 | 0 |
| **Percent of seats under Plan 2** | 100% | 0% |

*Fictional data created for practice. Simplified: equal-population precincts, and the same statewide vote under both plans. Remember: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).*

**A. Using Stimulus 1, identify the gerrymandering technique used to draw the district boundaries in Plan 2.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names cracking
- Describes cracking: Party B's voters split across districts so they're outnumbered

MODEL ANSWER

Plan 2 uses cracking: Party B's voters are split across all five districts so they are outnumbered in every one.

**Common mistake:** "Packing, because Party A packed the districts with its own voters." That mixes up the terms. Packing crowds the OTHER party's voters together. Plan 2 does the opposite: it spreads Party B's voters out so they lose everywhere.

**Why the model works:** It names the right term (cracking) and shows how it works: Party B's voters are split up and outnumbered everywhere.

**B. Using Stimulus 1, describe how the district boundaries in Plan 2 divide Party B's voters.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes the split: about 2 Party B precincts in each district
- Describes the result: Party B is outnumbered in every district

MODEL ANSWER

**T1-** Plan 2's district lines split Party B's cluster of voters into five pieces.

**E2-** Party B's 10 precincts are bunched in the center and south, but the lines slice that cluster like a pizza, so each district gets exactly 2 Party B precincts and 3 Party A precincts.

**A3-** Because Party B has only 2 of 5 precincts in every district, it is outnumbered 3 to 2 everywhere and cannot win a single district.

**Common mistake:** "Plan 2 divides Party B's voters into different districts." That just repeats the question. Describe HOW they're divided (2 Party B precincts in each district) and what that does (Party B is outnumbered everywhere).

**Why the model works:** It describes the split, gives map evidence (2 Party B precincts per district), and describes the result (outnumbered everywhere).

**C. Explain why the boundaries of congressional districts in the United States are redrawn every ten years.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the census as the trigger
- Explains the chain: seats shift between states, so lines are redrawn to keep districts equal

MODEL ANSWER

**T1-** District lines are redrawn every ten years because the U.S. Census counts the population every ten years, and populations shift.
**E2-** After each census, the House's 435 seats are reapportioned among the states by population; after 2020, Texas gained two seats and New York lost one.
**A3-** Because a state's population and seat count can change, it must redraw its districts (redistricting) so each has about the same number of people: one person, one vote.

**Common mistake:** "Because politicians want to change the lines to help their party win." That's why people GERRYMANDER, not why lines are redrawn every ten years. The required reason is the census and shifting population.

**Why the model works:** It names the trigger (the census), explains reapportionment with real examples, and explains why lines must change (equal population).

**D. Using both stimuli, explain how Plan 2 produces an election result that does not match the statewide vote.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses Stimulus 2: votes vs. seats (Party B: 40% of votes, 0% of seats)
- Uses Stimulus 1 for the cause: Party B has only 2 of 5 precincts in each district

MODEL ANSWER

**T1-** Plan 2 gives Party A every seat because it cracks Party B's voters so they are a minority in every district.
**E2-** Stimulus 2 shows Party B won 40% of the statewide vote but 0% of the seats under Plan 2, and Stimulus 1 shows why: each Plan 2 district has exactly 2 Party B precincts and 3 Party A precincts.
**A3-** Because Party B is outnumbered 3 to 2 in all five districts, its votes are wasted everywhere, so Party A turns 60% of the vote into 100% of the seats.

**Common mistake:** "Party A got 60% of the vote but won 100% of the seats." True, but it only restates Stimulus 2. The question asks HOW, so add Stimulus 1: Party B has only 2 of 5 precincts in every district.

**Why the model works:** It uses BOTH stimuli: the table for the mismatch (40% of votes, 0% of seats) and the map for the cause (2 of 5 precincts per district).

**E. Using Stimulus 2, explain ONE effect that a districting plan like Plan 2 could have on voters or on political representation.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses Stimulus 2: Party B wins 40% of the vote but 0 seats
- Explains an effect: no representation, wasted votes, lower turnout, safe seats, or polarization

MODEL ANSWER

**T1-** One effect of a plan like Plan 2 is that a large group of voters ends up with no representation, which can lower voter turnout.
**E2-** Stimulus 2 shows Party B won 40% of the statewide vote but 0% of the seats under Plan 2, so four in ten voters have no representative from the party they chose.
**A3-** Because Party B's voters lose in every district no matter how many show up, many may feel their votes don't count and stop voting.

**Common mistake:** "Gerrymandering is unfair to Party B." Too vague. It doesn't use the table or name an effect. Say what happens: Party B gets 40% of the vote but 0% of the seats, so its voters may stop voting.

**Why the model works:** It quotes Stimulus 2 (40% of votes, 0% of seats), names a specific effect (lower turnout), and explains how Plan 2 causes it.

**F. Explain ONE way that an independent redistricting commission OR a compactness requirement could reduce gerrymandering like the one in Plan 2.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the method: an independent commission or a compactness rule
- Explains how: takes line-drawing away from politicians who benefit, or bans odd shapes that split up voters

MODEL ANSWER

**T1-** An independent redistricting commission could reduce gerrymandering by taking map-drawing away from politicians.
**E2-** Arizona, Colorado, and Michigan created commissions of citizens from both major parties plus independents, who draw the maps instead of the state legislature.
**A3-** Because these mapmakers aren't trying to keep their own party in power, they have less reason to crack one party's voters the way Plan 2 cracks Party B.

**Common mistake:** "An independent commission would make redistricting fair." It names a fix but never explains HOW. Add the mechanism: the commission takes map-drawing away from politicians who benefit from cracking the other party's voters.

**Why the model works:** It names one fix, gives real examples, and explains how it reduces cracking (the mapmakers don't benefit from it).

**G. Explain how district boundaries like those in Plan 2 could act as a centrifugal force in a country.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Describes a divisive reaction: distrust, anger, feeling unrepresented, or protest
- Connects it to a centrifugal force: it divides people or weakens unity and loyalty

**MODEL ANSWER**

**T1-** Unfair district boundaries like Plan 2's can act as a centrifugal force because they make a large group of citizens lose trust in their government.
**E2-** Under Plan 2, Party B's voters cast 40% of the votes but win no seats, so they may feel cheated and protest or refuse to accept election results.
**A3-** When many citizens believe the system is rigged, tension between groups grows and loyalty to the government weakens, which pulls the country apart.

**Common mistake:** "A centrifugal force is something that divides a country." A correct definition, but it never connects to Plan 2. Explain how unfair districts make voters lose trust in government, and how that divides the country.

**Why the model works:** It explains the reaction (lost trust), uses the scenario (40% of the votes, no seats), and connects it to the definition (weaker loyalty).

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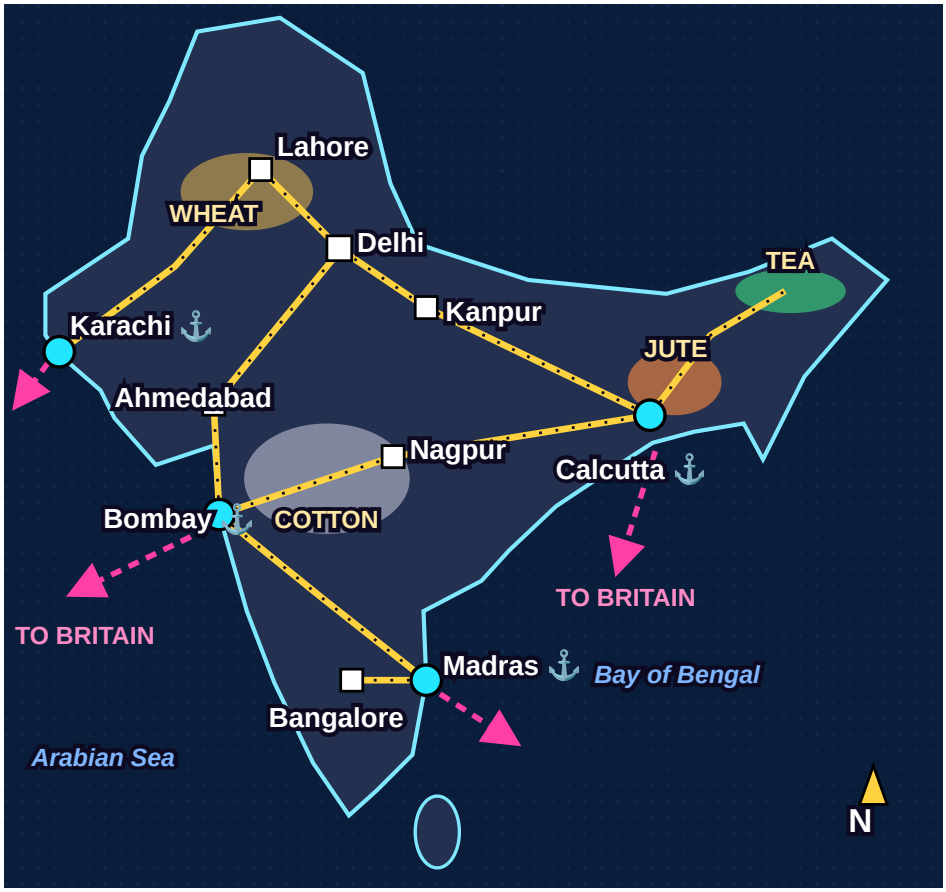
Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# Rails of the Raj

AP Human Geography · Units 4 · 7 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

This map shows major railway lines in British India around 1900, the port cities they reached, and regions that specialized in export crops.

**Railways, Ports, and Export Regions in British India, c. 1900**




Railway · Port city · Inland city · Wheat · Cotton · Jute · Tea

*Schematic map created for practice. Simplified, not to scale.*

**A. Identify the spatial pattern of the railway network shown on the map.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Lines link the interior to coastal ports
- Lines converge on a few major cities (e.g., Bombay, Calcutta)

MODEL ANSWER

The railway lines run from the interior of India and converge on a few coastal port cities, such as Bombay and Calcutta.

**Common mistake:** “The railroads are all over India.” Too vague. It doesn't say WHAT the pattern is, and 'all over' could describe any map.

**Why the model works:** It names a specific pattern (interior → coast) and points to real places on the map.

**B. Describe how the pattern of the railway network reflects Great Britain's economic goals in colonial India.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the goal: extracting raw materials or cash crops for Britain
- Links the rail pattern to moving goods from the interior to ports for export

MODEL ANSWER

**T1-** The railway pattern reflects Britain's goal of extracting raw materials from India.
**E2-** The lines mostly run from crop-growing regions in the interior, like cotton in the Deccan and jute in Bengal, to the ports of Bombay and Calcutta.
**A3-** This let raw materials move quickly to ships bound for factories in Britain, which is exactly what a colonial economy was built to do.

**Common mistake:** "Britain wanted to make India better by building trains." No economic goal and no map evidence. It never mentions raw materials or ports.

**Why the model works:** It names the goal (raw materials), uses map evidence (crop regions linked to ports), and connects the two.

**C. Explain how colonial rule contributed to the growth of port cities such as Bombay and Calcutta.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a colonial function: export/trade hub, rail terminus, or administrative center
- Explains growth through jobs, investment, or migration

MODEL ANSWER

**T1-** Colonial rule made Bombay and Calcutta grow because they became the gateways between India and Britain.
**E2-** The rail lines ended there, so these cities handled the exports leaving India and the British goods arriving, which concentrated jobs in shipping, warehouses, banking, and government.
**A3-** Those jobs pulled in rural migrants, and more people brought more businesses, so they became some of South Asia's largest cities.

**Common mistake:** "Bombay and Calcutta grew because they were big cities." Circular reasoning: they grew because they were big? There's no cause and no process.

**Why the model works:** It walks the whole chain: colonial trade role → jobs → migration → growth.

**D. Explain ONE way the colonial emphasis on export cash crops affected rural communities in South Asia.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names an effect: less food grown, famine, debt, poverty, or dependence on world prices
- Explains the cause-and-effect link to cash-crop farming

MODEL ANSWER

**T1-** The colonial push for export cash crops made rural families more vulnerable to hunger.
**E2-** Farmers were pressured to grow cotton and jute instead of food crops, so less land produced food and families depended on world prices they couldn't control.
**A3-** Because of this, when prices fell or a drought hit, people couldn't afford food, and drought years turned into deadly famines.

**Common mistake:** “Cash crops were bad for farmers.” It says 'bad' but not HOW. An Explain answer needs the cause-and-effect chain.

**Why the model works:** It names a specific effect (hunger and famine) and explains the chain that caused it.

**E. Describe how the economic relationship shown on the map illustrates Wallerstein's world-system theory.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Places Britain as the core and India as the periphery
- Describes the flow: raw materials out to the core, manufactured goods or profits kept by the core

MODEL ANSWER

**T1-** In world-system theory, Britain was the core and India the periphery.
**E2-** The map shows India sending low-value raw materials like cotton and jute to ports for export to Britain.
**A3-** Britain turned those materials into manufactured goods and kept the higher profits and industrial jobs, which is how the core grows wealthy from the periphery.

**Common mistake:** “Wallerstein's theory says there are core and periphery countries.” It defines the theory but never applies it to Britain, India, or the map.

**Why the model works:** It labels each country AND uses the map's flow of raw materials to show the relationship.

**F. Explain ONE way the colonial transportation network continues to affect patterns of economic development in South Asia today.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a present-day pattern: old ports remain top economic hubs, or coast/interior development is uneven
- Explains why: existing infrastructure keeps attracting investment, industry, and migrants

**MODEL ANSWER**

**T1-** The colonial transportation network still concentrates development in the old port cities.
**E2-** Because colonial railways tied the economy to a few ports, cities like Mumbai and Kolkata got a head start in infrastructure, industry, and finance.
**A3-** That infrastructure keeps attracting new businesses and migrants (cumulative causation), so development is still uneven, with many interior rural regions lagging behind.

**Common mistake:** "India is still affected by colonialism today." True, but it names no specific effect and doesn't explain how the railways cause it.

**Why the model works:** It names a present-day pattern (old ports still lead) and explains why (infrastructure keeps attracting investment).

**G. Explain ONE limitation of using this map to understand how colonialism affected people in South Asia.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- States a specific limit (no data on people, incomes, deaths, or experiences; a single time period; one scale)
- Explains why that prevents fully understanding colonialism's effects

**MODEL ANSWER**

**T1-** One limitation is that the map shows infrastructure, not people.
**E2-** It has no data on incomes, famine deaths, literacy, or how Indians experienced colonial rule.
**A3-** Because of that, it can't show who actually benefited or was harmed by the railways, so by itself it can't measure colonialism's human impact.

**Common mistake:** "The map is too small and hard to read." That's about the picture's quality, not its information. AP wants to know what content is missing.

**Why the model works:** It names specific missing data and explains which conclusion you can't reach without it.

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# Miracle Seeds

AP Human Geography · Unit 5 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

**No stimulus.** Since 1950 the world's population has more than tripled to over 8 billion, and farmers now grow enough food to feed everyone, yet hundreds of millions of people still go hungry. This FRQ asks how farming changed to feed a growing world, who got left out, and how to keep it going without wearing out the land.

**A. Identify ONE agricultural innovation associated with the Green Revolution.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names high-yield seed varieties (HYVs)
- Names another input: fertilizer, pesticides, irrigation, or machines

MODEL ANSWER

High-yield varieties (HYVs) of wheat and rice, like the short, sturdy wheat Norman Borlaug bred in Mexico.

**Common mistake:** “Farming got better.” That's a result, not an innovation. Name the actual new thing, like high-yield seeds or chemical fertilizer.

**Why the model works:** It names a specific innovation (high-yield varieties), the crops (wheat and rice), and a real example (Borlaug's wheat in Mexico).

**B. Describe ONE way the Green Revolution changed food production in a specific country, such as India or Mexico.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a change: higher yields, more food, or self-sufficiency
- Gives proof from a real country (India, Mexico...)

MODEL ANSWER

**T1-** The Green Revolution greatly increased how much food India's farms produced per acre.
**E2-** Starting in the mid-1960s, farmers in the Punjab region planted high-yield wheat with chemical fertilizer and irrigation, and India's wheat harvest about doubled in less than ten years.
**A3-** This turned India from a country that depended on U.S. wheat shipments into one that, by the late 1970s, grew enough grain to feed its own people.

**Common mistake:** “The Green Revolution made farming better in India.” “Better” is too vague. Say WHAT changed (much bigger wheat and rice harvests) and give proof, like India's wheat harvest doubling.

**Why the model works:** It names the change (more food per acre), proves it with a real place (Punjab, India's harvest doubling), and describes the result (India could feed itself).

**C. Explain ONE negative environmental consequence of the Green Revolution.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a harm: groundwater depletion, salty soil, pollution, or lost crop diversity
- Explains how Green Revolution farming causes it

MODEL ANSWER

**T1-** One negative environmental consequence of the Green Revolution is the depletion of groundwater.
**E2-** In India's Punjab region, high-yield rice and wheat need heavy irrigation, so farmers pump water from tube wells faster than rain can refill the aquifer, and the water table has dropped year after year.
**A3-** Because the farms depend on that water, a falling water table threatens to make the high yields impossible to keep up, putting future food production at risk.

**Common mistake:** "The Green Revolution was bad for the environment." That just repeats the question. Name a specific harm, like falling groundwater, and explain how the farming caused it.

**Why the model works:** It names a specific harm (groundwater depletion), uses a real place (Punjab), and explains the chain from thirsty crops to a falling water table.

**D. Explain ONE reason the Green Revolution benefited large farms more than small farms.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: the inputs were costly, and big farmers had more money or credit
- Explains how that helped large farms or hurt small ones

MODEL ANSWER

**T1-** The Green Revolution helped large farms more because its high-yield package was expensive, and wealthy farmers could afford it more easily.
**E2-** In India, the new seeds only paid off with bought fertilizer, pesticides, and irrigation pumps, and large landowners could get bank loans using their land as collateral, while small farmers often borrowed from moneylenders at high interest.
**A3-** So large farms earned bigger profits and bought more land, while many small farmers fell into debt and some sold their land, widening the gap between rich and poor farmers.

**Common mistake:** "Large farms have more space to grow crops." True of any farming. It doesn't explain what about the Green Revolution helped them: the costly inputs big farmers could afford.

**Why the model works:** It names the reason (an expensive package), gives proof (bank loans vs. moneylenders), and explains the result (profits for big farms, debt for small ones).

**E. Explain ONE reason the Green Revolution had less impact in sub-Saharan Africa than in South Asia.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: different staples, little irrigation, high costs, or poor roads
- Explains how that limited the benefits

MODEL ANSWER

**T1-** The Green Revolution had less impact in sub-Saharan Africa because its high-yield seeds were bred for crops and conditions that didn't match most African farms.
**E2-** Scientists focused on irrigated wheat and rice, but many farmers in sub-Saharan Africa grow staples like cassava, sorghum, and millet on rain-fed fields, and only a small share of the region's farmland is irrigated.
**A3-** Since the new seeds didn't fit Africa's main crops or its dependence on rainfall, yields there rose much less than in South Asia.

**Common mistake:** "Africa is a desert, so nothing grows there." Not true: most of sub-Saharan Africa is savanna, farmland, and rainforest. Give a real reason, like seeds bred for irrigated wheat and rice instead of cassava and millet.

**Why the model works:** It names a clear reason (seeds bred for other crops), gives proof (cassava, sorghum, millet; little irrigation), and explains why yields rose less.

**F. Explain ONE example of a sustainable farming practice and how it reduces environmental harm caused by agriculture.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a sustainable practice (no-till, crop rotation, terracing, organic, CSA...)
- Explains how it reduces a specific harm (erosion, water use, chemicals)

MODEL ANSWER

**T1-** One sustainable farming practice is conservation tillage, also called no-till farming.
**E2-** Instead of plowing the whole field each year, many farmers in places like the U.S. Midwest, Brazil, and Argentina plant seeds directly into the stalks and leaves left over from last year's crop.
**A3-** Because that leftover plant cover protects the ground from wind and rain, much less soil erodes away, which keeps fields fertile and keeps mud out of nearby rivers.

**Common mistake:** "Sustainable farming helps the environment because it is eco-friendly." That's circular: it just says sustainable is good for nature. Name ONE actual practice, like no-till or terracing, and explain HOW it reduces a specific harm.

**Why the model works:** It names a real practice (no-till), shows where it's used (the U.S. Midwest, Brazil, Argentina), and explains the cause → effect (plant cover → less erosion).

**G. Explain what a food desert is and ONE reason food deserts exist in urban or rural areas of more developed countries.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Defines a food desert: little access to affordable, healthy, or fresh food
- Explains a reason: stores avoid poor or sparse areas, or residents lack cars

**MODEL ANSWER**

**T1-** A food desert is an area where people have little access to affordable, healthy food because there are few or no full-service grocery stores nearby.
**E2-** In many low-income city neighborhoods, such as parts of Detroit and Chicago's South Side, supermarket chains closed stores and opened new ones in the suburbs, where customers had more money to spend.
**A3-** Because grocery companies locate where they expect the highest profits, residents without cars must rely on corner stores and fast food, which leads to poorer diets.

**Common mistake:** "A food desert is a place where there is no food, like the Sahara." That's the most common mix-up! A food desert usually has food (chips and soda at the corner store) but little affordable, healthy food. And you still need a reason it exists.

**Why the model works:** It defines the term (little access to affordable, healthy food), gives real places (Detroit, Chicago's South Side), and explains the cause (stores follow profits).

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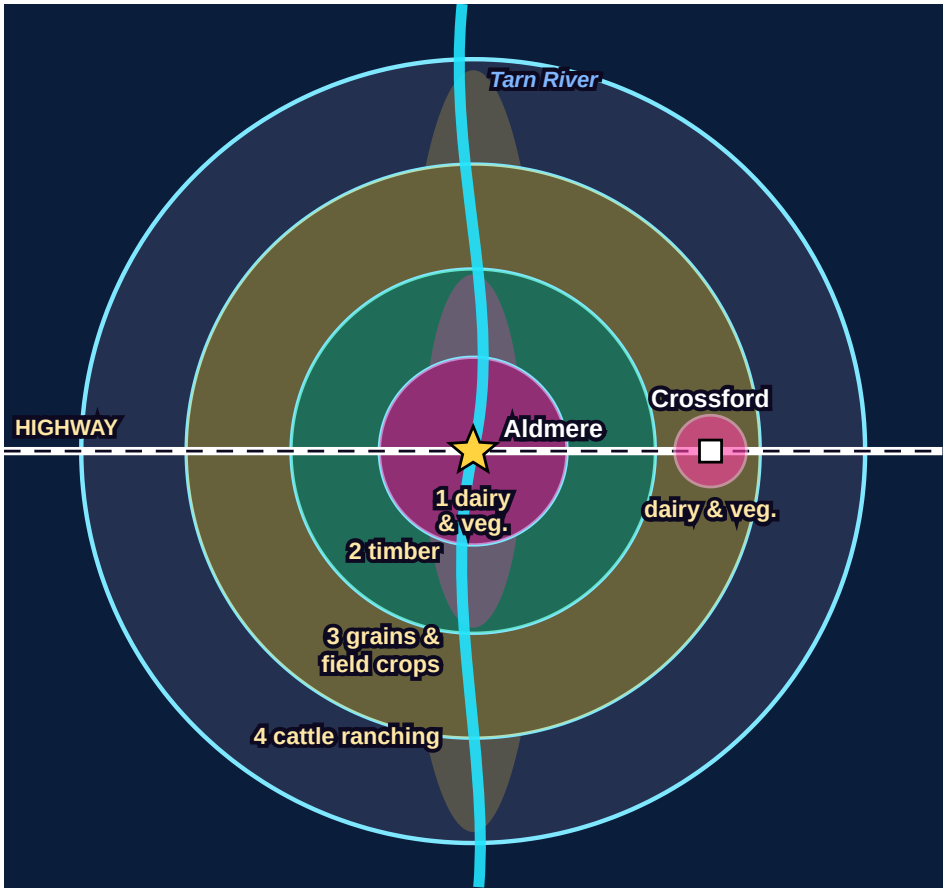
Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# Rings of Farmland

AP Human Geography · Unit 5 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

This diagram shows land use around a market city in a fictional farming region. Farmers sell their products in the city of Aldmere. A river flows through the region, and a highway crosses it from west to east.

**Agricultural Land Use Around Aldmere, a Fictional Market City**




1 Dairy & vegetables · 2 Timber / forest · 3 Grains & field crops · 4 Cattle ranching · River · Highway / town · Market city

*Diagram created for practice. Aldmere, Crossford, and the Tarn River are fictional. Simplified and not to scale; lighter shading shows where a ring is stretched.*

**A. Identify the agricultural activity found in ring 1, the ring closest to the market city.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Names dairy farming or vegetable/market gardening
- Describes it as perishable or intensive farming near the city

MODEL ANSWER

Ring 1 is dairy farming and market gardening (growing fresh vegetables) for the city.

**Common mistake:** "Farms." Every ring is farms. Identify means name the specific activity: dairy and vegetable (market) gardening.

**Why the model works:** It names the exact activity on the label (dairy and vegetables) and uses the geographer's term for it (market gardening).

**B. Explain why the agricultural products grown closest to the market are the most perishable or the most expensive to transport.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names the trait: perishable (spoils fast) or heavy/bulky products like milk, vegetables, or timber
- Explains the logic: transport cost rises with distance, so these farmers gain the most from being close and can outbid others for near-city land (bid-rent)

MODEL ANSWER

**T1-** Products that spoil quickly or are heavy to haul are grown closest to the market because transport cost rises with distance from the city.
**E2-** Milk and fresh vegetables can rot within a day or two of harvest, and timber is bulky and heavy to carry by wagon, while grain keeps for months and cattle can walk to market.
**A3-** Because dairy farmers and market gardeners lose the most money from a long trip, they gain the most from being near the city and can afford to pay (bid) more for the expensive land there, which pushes grain and ranching farther out.

**Common mistake:** "Farmers near the city grow what city people want to buy." That doesn't explain distance. City people buy bread and beef too, but grain and cattle come from far away. You need the transport-cost and spoilage logic.

**Why the model works:** It states the rule (transport cost rises with distance), gives specific products as proof (milk, vegetables, timber vs. grain and cattle), and explains the bidding result (perishable-product farmers outbid others for near-city land).

**C. Using the diagram, describe how the river changes the pattern of land use compared with the classic von Thünen model.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Describes the change with diagram evidence: rings (1 and 3) are stretched/elongated into ovals along the river instead of perfect circles
- Links it to cheaper or faster transport along the river (boats/barges lower transport cost)

**MODEL ANSWER**

**T1-** In the classic model the rings are perfect circles, but on this diagram the river stretches the rings into long ovals.
**E2-** The lighter shading shows ring 1 (dairy and vegetables) extending north and south of Aldmere along the Tarn River into the timber ring, and ring 3 (grains) extending along the river into the ranching ring.
**A3-** This happens because moving goods by boat along the river is cheaper and faster than hauling them by wagon, so a farm that is far away in miles but on the river is “close” in transport cost and can grow the same products as farms next to the city.

**Common mistake:** “The river gives the farms water, so farmers near the river can grow more crops.” That's irrigation, not the model. The question is about the SHAPE of the rings. Describe how they are stretched along the river and connect it to cheaper transport by boat.

**Why the model works:** It contrasts the classic circles with the stretched ovals, cites the exact rings and direction from the diagram, and explains the cause (cheaper boat transport makes river farms “closer” in cost).

**D. Explain how the secondary market town of Crossford along the highway shows how modern transportation changes the von Thünen model.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Identifies what Crossford shows: a second market with its own mini dairy/vegetable zone (ring), breaking the one-market assumption
- Explains the role of modern transport: highways/trucks make travel faster and cheaper, so markets and rings form in new places

**MODEL ANSWER**

**T1-** Crossford shows that modern transportation breaks the model's assumption of a single market with one set of rings.
**E2-** On the diagram, a small dairy-and-vegetable zone surrounds Crossford where the highway crosses ring 3, forming a miniature ring 1 inside the grain zone.
**A3-** Because trucks on the highway move goods quickly and cheaply, Crossford grew into a second market of its own, so the region now has overlapping sets of rings instead of one neat set of circles around Aldmere.

**Common mistake:** “The highway makes it easier for people to drive to Aldmere.” That's about commuters, not land use. The question wants what Crossford shows about the model: a second market with its own mini ring, created because trucks on the highway cut transport cost.

**Why the model works:** It names the broken assumption (one market), uses diagram evidence (the mini dairy zone around Crossford on the highway), and explains the cause (fast, cheap trucking creates a second market and overlapping rings).

**E. Explain ONE way refrigerated trucking and global shipping have weakened the von Thünen pattern at the national or global scale.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names the technology: refrigeration/cold chain, refrigerated trucks, container ships, air freight, highways
- Explains the effect with scale: perishables are grown far away (e.g., Chile, Mexico, California) so the near-city rings weaken at the national or global scale

**MODEL ANSWER**

**T1-** Refrigeration has weakened the model because perishable products no longer have to be grown next to the city that eats them.

**E2-** Refrigerated trucks and container ships keep produce fresh for weeks, so a U.S. supermarket in January sells grapes from Chile and tomatoes from Mexico that traveled thousands of miles.

**A3-** At the national and global scale, this means the ring 1 logic breaks down: the dairy-and-vegetable zone is not a ring around each city but far-off regions like California, Florida, or another country, so the whole world acts as one giant hinterland.

**Common mistake:** “Trucks are faster than horses, so people can sell more.” That's true but doesn't explain the model. Name the technology (refrigeration), give an example (Chilean grapes in January), and explain how it breaks the near-city ring at the national or global scale.

**Why the model works:** It names the technology (refrigerated trucks and container ships), gives a concrete example with distance (Chile, Mexico, thousands of miles), and explains the effect at a named scale (national/global rings dissolve).

**F. Describe ONE environmental consequence of intensive commercial agriculture in ring 1 or ring 3 of the diagram.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a specific environmental consequence: fertilizer/pesticide/manure runoff, water pollution, aquifer or groundwater depletion, soil salinization, soil erosion, or loss of soil fertility
- Describes how it happens or what it does to the environment (cause → result)

**MODEL ANSWER**

**T1-** One environmental consequence of intensive farming in ring 3 is water pollution from fertilizer runoff.

**E2-** Grain farmers spread large amounts of nitrogen fertilizer on their fields, and when it rains the extra fertilizer washes off the fields and flows into the Tarn River.

**A3-** The nitrogen feeds huge algae blooms downstream that use up the oxygen in the water and kill fish, creating a "dead zone" like the one at the mouth of the Mississippi River in the Gulf of Mexico.

**Common mistake:** "Farming in ring 3 uses up a lot of land." Using land isn't an environmental consequence by itself. Name a specific harm (fertilizer runoff, aquifer depletion, salinization) and describe how it happens.

**Why the model works:** It names a specific consequence (fertilizer runoff), describes the mechanism (rain washes nitrogen into the river), and gives the result with a real example (algae, dead zones, Gulf of Mexico).

**G. Explain ONE limitation of using this model to understand agriculture in a less developed country today.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names an assumption that doesn't fit: subsistence (not commercial) farming, no single market / many small markets, export crops for ports, rugged terrain or varied climate, land tenure (communal land, plantations, landless farmers)
- Explains why that makes the model misleading there

**MODEL ANSWER**

**T1-** One limitation is that the model assumes commercial farmers who sell everything to a single market city, which does not describe many farmers in less developed countries.
**E2-** In rural parts of countries like Ethiopia or Bolivia, many families practice subsistence agriculture, growing maize, teff, or potatoes mainly to feed themselves, often on steep hillsides or communal land rather than a flat plain with one market.
**A3-** Because those farmers are not choosing crops based on transport cost to a central city, distance to the market does not sort their land use into rings, so the model would predict the wrong pattern and miss what actually shapes their farming, such as rainfall, terrain, and land ownership.

**Common mistake:** “The model is old, so it doesn't work anymore.” Being old isn't a limitation by itself; the transport-cost logic still works in many places. Name a specific assumption that fails in a less developed country (subsistence farming, no single market, terrain, land tenure) and explain why that matters.

**Why the model works:** It names the failing assumption (commercial farmers and a single market), gives specific detail (subsistence farming on hillsides and communal land in Ethiopia or Bolivia), and explains the consequence (distance doesn't sort land use, so the model predicts wrongly).

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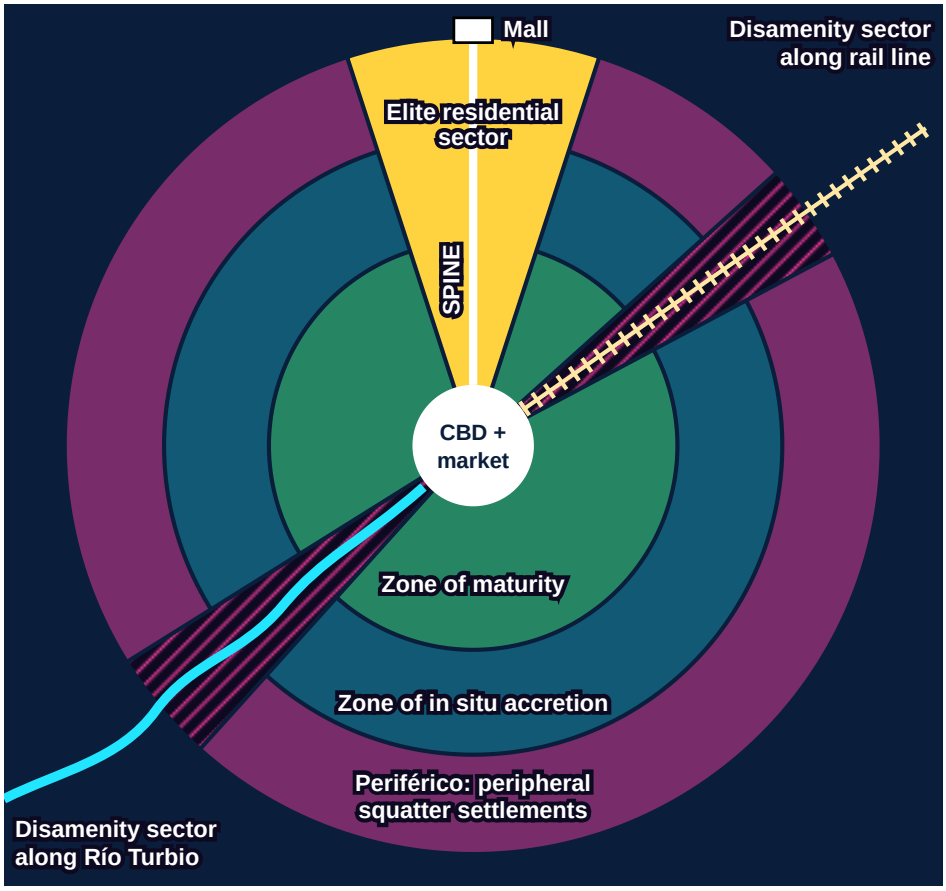
# City of Contrasts

AP Human Geography · Unit 6 · Two stimuli (like Question 3) · 7 points · Teacher key · GeoArcade FRQ Tutor

Puerto Alba is the largest city of Valdoria, an invented Latin American country. Like many cities in developing countries, it has grown fast as people pour in from the countryside, and its rich and poor neighborhoods are arranged very differently from a typical U.S. city's.

**Stimulus 1: Model of Puerto Alba, a Fictional Latin American City · Stimulus 2: The Four Largest Cities of Valdoria, 2000 and 2025**

**Stimulus 1: Model of Puerto Alba, a Fictional Latin American City**




CBD & market · Spine & elite residential sector · Zone of maturity · Zone of in situ accretion · Periférico (squatter settlements) · Disamenity sector · River · Rail line

*Created for practice, based on the Griffin-Ford model. Puerto Alba and the Río Turbio are fictional. Not to scale.*

**Stimulus 2: The Four Largest Cities of Valdoria, 2000 and 2025**

| Rank | City | Population, 2000 | Population, 2025 | Rank-size rule prediction, 2025 (largest ÷ rank) |
|---|---|---|---|---|
| **1** | Puerto Alba | 5,400,000 | 9,600,000 | 9,600,000 |
| **2** | San Lucero | 900,000 | 1,300,000 | 4,800,000 |
| **3** | Aguasol | 650,000 | 900,000 | 3,200,000 |
| **4** | Tres Cerros | 480,000 | 650,000 | 2,400,000 |

*Valdoria, its cities, and all data are fictional, created for practice. Rank-size rule: city number n should have 1/n the population of the largest.*

*Class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).*

**A. Identify the zone in Stimulus 1 where most of Puerto Alba's squatter settlements are located.** IDENTIFY

**EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:**

- Names the periférico (the zone of peripheral squatter settlements)
- Locates it as the outermost ring at the city's edge

**MODEL ANSWER**

The periférico: the zone of peripheral squatter settlements, the outermost ring at the edge of the city.

**Common mistake:** “Slums.” That's a kind of housing, not a zone (the disamenity sectors have slums too). Name the zone: the periférico.

**Why the model works:** It names the zone (the periférico) and says where it is (the outermost ring).

**B. Using Stimulus 1, describe the spine and the elite residential sector in Puerto Alba.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Describes the spine: a commercial boulevard out from the CBD
- Describes the elite sector: wealthy homes along the spine

**MODEL ANSWER**

**T1-** The spine is a commercial boulevard running out from the CBD, and the elite residential sector is the wedge of wealthy homes on both sides of it.
**E2-** In Stimulus 1, the spine runs north from the CBD to a mall at the edge, and the gold elite sector follows it through all three rings.
**A3-** Because the spine has the city's best shops, offices, and parks, the richest families live along it, so wealth forms a wedge, not a ring.

**Common mistake:** “The spine is on the north side of the city, and the elite residential sector is next to it.” That only gives names and places. Describe what they ARE: a commercial boulevard, lined by the wealthiest homes.

**Why the model works:** It defines the spine, says who lives in the elite sector, and uses diagram details (north to a mall).

**C. Explain ONE reason why wealthy residents of many Latin American cities live near the CBD and along the spine, unlike in many U.S. cities.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a reason: services, amenities, jobs, safety, colonial tradition, or status
- Explains the contrast: the edge lacks services, while U.S. elites moved to suburbs

**MODEL ANSWER**

**T1-** Wealthy residents live near the CBD and spine because that is where the best services and amenities are.
**E2-** The center has reliable water, electricity, police, and the best shops and parks, while the periférico at the edge often lacks piped water and sewers.
**A3-** Because the edge has no services, living near the center brings comfort, safety, and status, the opposite of the U.S., where cars and highways let the wealthy move out to suburbs.

**Common mistake:** "Rich people like to live in nice areas." Everyone does! Say WHAT makes the center nice (like better services) and contrast it with the edge or the U.S.

**Why the model works:** It names a reason (services), gives proof (water, police), and explains the U.S. contrast.

**D. Using Stimulus 2, explain whether Puerto Alba is a primate city. Support your answer with data from the table.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Uses table data (9.6 vs. 1.3 million, or the 4.8 million prediction)
- Explains that it IS primate: more than twice the next city

**MODEL ANSWER**

**T1-** Yes, Puerto Alba is a primate city because it is far more than twice as large as Valdoria's second-largest city.
**E2-** Stimulus 2 shows Puerto Alba has 9,600,000 people, more than seven times San Lucero's 1,300,000, while the rank-size rule predicts 4,800,000 for San Lucero.
**A3-** Because every smaller city falls far below its prediction, Valdoria's people, jobs, and investment are concentrated in one dominant city.

**Common mistake:** "Yes, Puerto Alba is a primate city because it is the largest city in Valdoria." Every country has a largest city, and there's no data. Prove it: 9,600,000 vs. 1,300,000.

**Why the model works:** It answers yes, proves it with table numbers, and explains what primacy means.

**E. Explain ONE cause of the rapid growth of squatter settlements on the edges of large cities in developing countries.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: rural-to-urban migration, population growth, or no affordable housing
- Explains the link: newcomers can't afford housing, so they build their own

MODEL ANSWER

**T1-** One cause is rapid rural-to-urban migration.
**E2-** Push factors like farm mechanization and rural poverty, plus pull factors like jobs and schools, send huge numbers of migrants to cities like Puerto Alba, which grew from 5,400,000 to 9,600,000 people since 2000.
**A3-** Because the city can't build affordable housing as fast as newcomers arrive, and most can't afford formal rents, many build their own homes on unused land at the edge, so squatter settlements spread.

**Common mistake:** "Because people want to live in the city." It doesn't say who is coming or why they squat. Name the cause (rural-to-urban migration) and the link (no affordable housing).

**Why the model works:** It names the cause, gives proof (push and pull factors, growth), and explains the link to housing.

**F. Using both stimuli, explain ONE challenge a primate city like Puerto Alba faces in providing services such as clean water, sanitation, or transportation to the periférico.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Uses Stimulus 1: the periférico is at the outer edge, far from the CBD
- Uses Stimulus 2: a primate city that grew fast
- Explains the service challenge and its effect

MODEL ANSWER

**T1-** One challenge is that Puerto Alba is growing faster than it can extend water pipes and sewers to its edge.
**E2-** Stimulus 2 shows Puerto Alba is a primate city that grew from 5,400,000 to 9,600,000 people since 2000, and Stimulus 1 shows the newest squatter settlements form the periférico, farthest from the CBD.
**A3-** As the periférico spreads onto unplanned land, pipes must stretch ever farther, so many residents go without clean water and sanitation, which spreads diseases like cholera.

**Common mistake:** "Puerto Alba has a lot of people, so it is crowded." It uses neither stimulus and names no service. Use both: fast growth (Stimulus 2) plus an edge far from the CBD (Stimulus 1).

**Why the model works:** It uses both stimuli (growth, the edge location), names a service, and explains the effect (disease).

**G. Explain ONE way a city government could improve living conditions in informal settlements like those in Puerto Alba's periférico.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a strategy: upgrading, services, land titles, site-and-service, or BRT
- Explains how it improves living conditions

**MODEL ANSWER**

**T1-** A city government could upgrade informal settlements in place by adding basic services and giving residents legal titles to their land.
**E2-** For example, Rio de Janeiro's Favela-Bairro program paved streets, installed water and sewer lines, and built community centers in existing favelas instead of bulldozing them.
**A3-** Clean water and sewers reduce disease, and families who securely own their land invest in improving their homes, which raises living conditions without forcing anyone to move.

**Common mistake:** “The government should get rid of the slums.” Clearing slums destroys homes and pushes people elsewhere. Name a strategy that helps and explain its effect.

**Why the model works:** It names a strategy, gives a real example (Favela-Bairro), and explains how lives improve.

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Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# Brookhaven Spreads Out

AP Human Geography · Unit 6 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Brookhaven is an invented U.S. metropolitan area of about one million people. Like many American metros, its population has shifted outward since 1970, while a few old neighborhoods near downtown have recently become popular, and expensive, again.

**Brookhaven Metropolitan Area: Population, Housing, and Commuting by Zone, 1970 and 2020**

| Zone | Population 1970 | Population 2020 | Median home price 2020 | Percent of workers who drive alone to work | Percent of residential land zoned single-family only |
|---|---|---|---|---|---|
| **Central city (inside the old core)** | 520,000 | 445,000 | $395,000 | 55% | 25% |
| **Inner suburbs** | 310,000 | 325,000 | $265,000 | 78% | 62% |
| **Outer suburbs / exurbs** | 95,000 | 300,000 | $380,000 | 90% | 85% |
| **Metro total** | 925,000 | 1,070,000 | $340,000 | 72% | 71% |

*Note: The central city's population fell to 388,000 in 2000, then grew again after 2010, mostly in two renovated neighborhoods near downtown (Millside and the Canal District), where the 2020 median home price was $610,000. Brookhaven and all of its data are invented for practice. Class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).*

**A. Identify the process shown by the growth of the outer suburbs and exurbs between 1970 and 2020.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Suburbanization or urban sprawl (also accept decentralization, exurbanization, outward growth)

MODEL ANSWER

Suburbanization, or urban sprawl: population spreading outward from the central city into low-density outer suburbs.

**Common mistake:** “Urbanization.” Urbanization means a bigger share of people living in cities overall. This table shows the center LOSING people while the far edge triples. That specific outward movement is suburbanization (sprawl).

**Why the model works:** It names the specific process (suburbanization/sprawl) and ties it to what the table shows: outward, low-density growth at the edge.

**B. Using the table, describe the relationship between distance from the center of Brookhaven and the percent of workers who drive alone to work.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Uses the table's numbers (55%, 78%, 90%)
- States the direction of the relationship: farther out means a higher drive-alone share (or closer in means lower)

**MODEL ANSWER**

**T1-** The farther a zone is from the center of Brookhaven, the higher the percent of workers who drive alone to work.
**E2-** In the central city only 55% drive alone, in the inner suburbs it rises to 78%, and in the outer suburbs and exurbs it reaches 90%.
**A3-** This fits because the outer zones are low-density and built around the car: homes are far from jobs and shops and there is little bus or rail service, while the center has transit, sidewalks, and jobs close together, so more people can walk, bike, or ride.

**Common mistake:** “Most people in Brookhaven drive to work alone.” True, but it's one fact about the whole metro. The question asks how the drive-alone share CHANGES with distance: compare the three zones and quote their numbers.

**Why the model works:** It states the direction (farther out = more driving alone), quotes all three numbers, and explains the pattern with density and transit.

**C. Explain ONE cause of the growth of Brookhaven's outer suburbs shown in the table.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a cause: cars/highways, cheaper land, government mortgage policy (FHA/VA loans), white flight, schools, more space, lower taxes or crime
- Explains how that cause produced growth at the edge (made moving out possible, cheaper, or attractive)

**MODEL ANSWER**

**T1-** One cause of the outer suburbs' growth is the spread of cars and highways.
**E2-** After the Interstate Highway System began in 1956, most families owned a car, so a worker could live 20 or 30 miles from downtown on cheap land and still commute to a job in the city.
**A3-** Because developers could build bigger, newer houses on inexpensive land at the edge and highways made the daily trip possible, families moved outward, and Brookhaven's outer suburbs tripled from 95,000 to 300,000 people while the center lost population.

**Common mistake:** “People wanted to live in the suburbs.” That's a wish, not a cause. What made it POSSIBLE to live far from downtown? Name something specific (highways and cars, cheap land, mortgages) and explain how it led people to move.

**Why the model works:** It names a specific cause (cars and highways), gives dated proof (the 1956 Interstate system), and explains the chain all the way to the table's numbers.

**D. Explain how the zoning data in the table could contribute to high housing prices in Brookhaven.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Identifies the constraint: single-family-only zoning blocks apartments and denser housing, limiting the number of homes (supply)
- Explains the link to price: limited supply plus demand leads to competition and higher prices

**MODEL ANSWER**

**T1-** Single-family-only zoning limits how many homes can be built, which raises prices.
**E2-** In Brookhaven, 85% of the outer suburbs' residential land and 71% of the metro's allows only one detached house per lot, so apartments, duplexes, and townhouses that could fit many more families are illegal on most of the land.
**A3-** Because the metro added about 145,000 people but zoning keeps the supply of homes small, buyers compete for the few houses available and bid prices up, which helps explain the $380,000 median in the outer suburbs.

**Common mistake:** “Rich people live in the outer suburbs, so houses cost more there.” That describes who lives there, not how ZONING works. The question is about the single-family-only percentage: explain how limiting what can be built limits supply and raises prices.

**Why the model works:** It names the constraint (single-family-only zoning limits supply), quotes the table (85%, 71%), and walks through supply, demand, and price.

**E. Explain how the recent growth and high home prices in some of Brookhaven's central-city neighborhoods could be an example of gentrification, including ONE effect on long-time residents.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Describes gentrification: wealthier or younger people moving into and renovating older central neighborhoods, raising values
- Gives one effect on long-time residents: displacement, unaffordable rent or taxes, loss of community or culture

MODEL ANSWER

**T1-** The comeback of a few central-city neighborhoods looks like gentrification: wealthier newcomers moving into older, lower-income areas near downtown.
**E2-** Brookhaven's central city shrank from 520,000 to 388,000 by 2000, but since 2010 it has grown again mostly in the renovated Millside and Canal districts, where the median home price is $610,000, far above the inner suburbs' $265,000.
**A3-** As investors and young professionals renovate old homes, rents and property taxes rise, so long-time lower-income residents, especially renters, can be displaced because they can no longer afford to stay in their own neighborhood.

**Common mistake:** “The city is growing again because it is popular now.” 'Popular' isn't a process. Say WHO is moving in and WHY (wealthier people buying and renovating cheap old housing near downtown), then give the effect on the people already there.

**Why the model works:** It defines gentrification, anchors it in the note ($610,000 Millside and Canal districts, growth since 2010), and gives a clear effect: displacement by rising rents and taxes.

**F. Describe ONE smart-growth or New Urbanism policy Brookhaven could adopt to reduce sprawl, and how it would work.** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a smart-growth or New Urbanism policy: urban growth boundary, greenbelt, transit-oriented development, mixed-use zoning, infill or brownfield redevelopment, denser housing, walkable design
- Describes how it works to reduce sprawl (keeps growth inside, near transit, denser, less driving, protects land at the edge)

**MODEL ANSWER**

**T1-** Brookhaven could use transit-oriented development (TOD) to pull growth inward instead of outward.
**E2-** The city would rezone the land within about a half mile of its rail and bus stations for apartments, townhouses, shops, and offices, the way Arlington, Virginia built dense, walkable neighborhoods around its Metro stations.
**A3-** Because thousands of new homes would be built inside the existing metro next to transit, fewer families would need a new house on farmland at the edge, and more workers could live where the drive-alone share is 55% rather than 90%, which reduces sprawl and traffic.

**Common mistake:** “Build more roads so people can get to work faster.” More roads is the opposite of smart growth: wider highways make living even farther out easier, which causes MORE sprawl. Pick a policy that keeps growth inside the city.

**Why the model works:** It names a recognized policy (TOD), describes exactly what changes (rezoning near stations) with a real example, and connects it to less sprawl and less driving.

**G. Explain ONE limitation of using data for Brookhaven's three large zones, or the metro total, to understand its housing or transportation problems.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a limitation: zone-level or metro-level averages hide variation within zones or neighborhoods, or city/suburb boundaries are defined differently, or rent and income data are missing
- Explains why it misleads or what data would be needed instead

**MODEL ANSWER**

**T1-** One limitation is scale: each row averages hundreds of thousands of people, so it hides big differences between neighborhoods.
**E2-** The central city's median home price is listed as $395,000, but the note says the renovated Millside and Canal districts are at $610,000, which means other central neighborhoods must be far cheaper; likewise the 55% drive-alone figure lumps transit-rich downtown together with car-dependent parts of the city.
**A3-** Because the zone averages blur these differences, a planner could not tell where housing is actually unaffordable or where buses are missing, and would need neighborhood or census-tract data to target the real problem.

**Common mistake:** “The data is from 2020, so it is out of date.” 2020 is the most recent census year, and the question is about the SIZE of the areas, not the date. Explain what three giant zones hide, such as differences between neighborhoods.

**Why the model works:** It names the limitation (scale and averages), proves it with the $610,000 vs. $395,000 gap, and explains how it could mislead a planner.

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Scoring guide written for the GeoArcade FRQ Tutor. Award each part independently, 1 point each.

# The Factory Floor

AP Human Geography · Unit 7 · No stimulus (like Question 1) · 7 points · Teacher key · GeoArcade FRQ Tutor

**No stimulus.** Every factory sits where it does for a reason: close to its raw materials, its customers, or cheap workers. When those costs change, whole industries move, and the places they leave behind have to find a new way to make a living.

**A. Identify ONE of the three location factors in Alfred Weber's least-cost theory of industrial location.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Transportation costs
- Labor costs
- Agglomeration

MODEL ANSWER

Transportation costs: the cost of moving raw materials to the factory and finished products to the market.

**Common mistake:** "Good weather and cheap land." Those can matter to real companies, but they aren't Weber's factors. Name one of his three: transportation, labor, or agglomeration.

**Why the model works:** It names one of Weber's three factors exactly and adds a few words showing what it means.

**B. Describe why a bulk-reducing (weight-losing) industry, such as copper smelting, usually locates near its source of raw materials.** DESCRIBE

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Says the finished product weighs less than the raw materials, because waste is removed
- Describes the payoff: processing near the mine cuts transportation (shipping) costs

MODEL ANSWER

**T1-** A bulk-reducing industry locates near its raw materials because its finished product weighs much less than its inputs.
**E2-** Copper ore is usually about 1 percent copper or less, so mills and smelters near mines in places like Arizona and Chile remove tons of waste rock to get a little metal.
**A3-** Processing near the mine means only the lighter copper gets shipped, not heavy waste rock, which keeps transportation costs low.

**Common mistake:** "Copper smelters are built near copper mines because that is where the copper is." That's circular. Describe WHY: the ore loses most of its weight in processing, so shipping it unprocessed wastes money.

**Why the model works:** It names the idea (the product weighs far less than the ore), gives proof (about 1 percent copper), and describes the payoff (cheaper shipping).

**C. Explain ONE reason the Industrial Revolution began in Great Britain in the 1700s.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a reason: coal, iron, capital, colonies, rivers/canals, workers, inventions, or political stability
- Explains how it helped industry start: powered machines, paid for factories, or moved goods cheaply

MODEL ANSWER

**T1-** One reason the Industrial Revolution began in Great Britain is that it had large deposits of coal and iron ore.

**E2-** Coal fueled the steam engines James Watt improved in the 1760s and 1770s, iron built the machines, and canals like the Bridgewater Canal (1761) moved coal cheaply to Manchester.

**A3-** Because fuel and metal were cheap and close by, British owners could build steam-powered factories before any other country.

**Common mistake:** "The Industrial Revolution started in Britain because Britain was rich and powerful." Too vague. Name the specific advantage (coal and iron, colonial trade, canals) and explain how it helped factories start.

**Why the model works:** It names an advantage (coal and iron), gives proof (Watt's steam engines, the Bridgewater Canal), and explains how it let factories start there first.

**D. Explain ONE cause of deindustrialization in the Rust Belt of the United States since the 1970s.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Names a cause: automation, offshoring to lower-wage countries, or foreign competition
- Explains how it cut Rust Belt jobs: lower costs elsewhere, fewer workers needed, plants closed or moved

MODEL ANSWER

**T1-** One cause of deindustrialization in the Rust Belt is offshoring: companies moved factory work to countries where labor costs much less.

**E2-** After NAFTA (1994) and China's entry into the World Trade Organization (2001), many makers of auto parts and appliances opened plants in Mexico and China, where wages were far lower than in Ohio or Michigan.

**A3-** Because the same products could be made more cheaply abroad, companies closed or shrank Rust Belt plants, which led to the loss of hundreds of thousands of factory jobs.

**Common mistake:** "The factories got old and rusty." That describes the Rust Belt, not its cause. Name a real cause (automation, offshoring, or foreign competition) and explain how it cost jobs.

**Why the model works:** It names a cause (offshoring), gives specific proof (NAFTA, China joining the WTO), and explains how that cost Rust Belt jobs.

**E. Using ONE real Rust Belt city as an example, explain ONE economic or social effect of deindustrialization on that city.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names an effect: job loss, population loss, a shrinking tax base, vacant buildings, or a shift to service jobs
- Uses a real Rust Belt city as the example
- Explains how losing factories caused the effect

**MODEL ANSWER**

**T1-** One effect of deindustrialization on Detroit, Michigan, was a huge loss of population, which shrank the city's tax base.

**E2-** As auto plants closed or moved away, workers lost their jobs and left: Detroit fell from about 1.85 million people in 1950 to about 640,000 in 2020, leaving tens of thousands of houses vacant.

**A3-** Because far fewer people and businesses paid taxes, the city struggled to pay for streetlights and police, which helped push Detroit into bankruptcy in 2013.

**Common mistake:** “Detroit became a bad place to live.” That's an opinion, not a specific effect. Name a real effect (like population loss) and explain how closing the auto plants caused it.

**Why the model works:** It names the effect (population loss, a shrinking tax base), uses a real city with real numbers (Detroit), and explains the chain to bankruptcy.

**F. Explain how microloans can promote economic development, especially for women in developing countries.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Says what microloans do: small loans that let poor women shut out of banks start small businesses
- Explains the payoff: income for her family, children's schooling and health, or empowerment

**MODEL ANSWER**

**T1-** Microloans promote development by lending small amounts to poor people, especially women, who are shut out of regular banks because they have no collateral.

**E2-** Bangladesh's Grameen Bank lends mostly to women, who use the money to buy a milk cow, a sewing machine, or a mobile phone to rent to neighbors, then repay from their sales.

**A3-** Because she now earns her own income, a borrower can spend more on her children's food, health, and schooling and gains a bigger voice in family decisions, which reduces poverty and empowers women.

**Common mistake:** “Microloans give people money.” That's true of every loan. Explain what the money is FOR (starting a small business) and how that leads to development.

**Why the model works:** It explains what the loan does (money for a small business), gives a real example (Grameen Bank), and explains the payoff (income for her children and a bigger voice).

**G. Explain ONE way ecotourism can support sustainable development, and explain ONE drawback of ecotourism.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Explains how ecotourism brings money or jobs from visitors who come to see nature
- Connects it to sustainability: gives people a reason to protect nature for the future
- Explains a drawback: crowds damage nature, travel pollution, or profits leaving the community

**MODEL ANSWER**

**T1-** Ecotourism can support sustainable development because it turns protecting nature into a source of jobs and income.
**E2-** Costa Rica protects about a quarter of its land in parks and reserves, and tourists pay to see its cloud forests, creating jobs for local guides and lodge owners; in Rwanda, visitors pay $1,500 each to see mountain gorillas.
**A3-** Because living forests and gorillas earn money year after year, locals have a reason to protect them for future generations; however, one drawback is that crowds of visitors can damage the fragile places they come to see.

**Common mistake:** “Ecotourism is good because people like to see animals.” Liking animals isn't a development reason. Explain how tourist money makes protecting nature pay, and add a drawback.

**Why the model works:** It explains how ecotourism earns money (jobs, permits), connects it to sustainability (people protect what keeps paying), and adds a drawback.

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# Measuring Up

AP Human Geography · Unit 7 · One stimulus (like Question 2) · 7 points · Teacher key · GeoArcade FRQ Tutor

Geographers use many indicators to measure development, because no single number tells the whole story. The table shows six indicators for four countries.

**Selected Development Indicators for Four Countries, 2023**

| Country | GNI per capita (PPP, US$) | HDI (0–1) | Gender Inequality Index (0–1, lower is better) | Workers in agriculture (%) | Workers in services (%) | Exports from one commodity (%) |
|---|---|---|---|---|---|---|
| **Ardena** | 48,200 | 0.93 | 0.08 | 2 | 78 | 9 (machinery) |
| **Kovar** | 14,100 | 0.76 | 0.35 | 18 | 55 | 20 (electronics) |
| **Sulmani** | 2,100 | 0.49 | 0.58 | 62 | 25 | 15 (coffee) |
| **Tessar** | 31,400 | 0.74 | 0.50 | 6 | 60 | 85 (petroleum) |

*Note: The countries and data are invented for practice. Remember the class directions: label each part, and write Describe and Explain parts as T1- (your reason in general words), E2- (the specific proof), A3- (why it matters).*

**A. Identify the country in the table that is most likely in Rostow's stage of high mass consumption.** IDENTIFY

EARNS THE POINT IF THE ANSWER DOES ANY ONE OF THESE:

- Ardena

MODEL ANSWER

Ardena.

**Common mistake:** “Tessar, because it has a high GNI per capita.” GNI alone is the trap in this table. Tessar's HDI is the lowest of the four and 85% of its exports are one raw material, so it isn't a mature, diversified economy. Check HDI and the services column too.

**Why the model works:** It names the one country whose numbers all point to the top stage: highest income, highest HDI, almost no farmers, and most workers in services.

**B. Using the table, describe the relationship between the percent of workers in agriculture and the Human Development Index (HDI).** DESCRIBE

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- States the direction: higher percent in agriculture goes with lower HDI (inverse/negative relationship)
- Uses the table as proof: names countries or quotes numbers (e.g., Sulmani 62% / 0.49 vs. Ardena 2% / 0.93)

**MODEL ANSWER**

**T1-** The table shows an inverse relationship: the higher the percent of workers in agriculture, the lower the HDI.

**E2-** Sulmani has 62% of its workers in agriculture and the lowest HDI (0.49), while Ardena has only 2% in agriculture and the highest HDI (0.93).

**A3-** This fits the idea that as countries develop, workers shift out of farming into factories and services, which raises incomes, education, and life expectancy.

**Common mistake:** "Agriculture and HDI are related." Related how? A relationship has a direction (as one goes up, the other goes down), and you need numbers from the table as proof.

**Why the model works:** It states the direction (inverse), proves it with two rows and their numbers, and explains why farming and HDI move in opposite directions.

**C. Using the table, explain why a country can have a high GNI per capita but a lower HDI than its income would predict.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a reason: wealth is concentrated or unequal, money comes from one resource (oil), low spending or corruption, or gender inequality
- Explains the link to HDI: money doesn't reach education, health, or life expectancy for most people

**MODEL ANSWER**

**T1-** A country can have a high GNI per capita but a lower HDI when its wealth is concentrated in a few hands and not spent on people.

**E2-** Tessar earns $31,400 per person on paper, but 85% of its exports are petroleum, so the money flows to the government and a small elite rather than into schools and hospitals; its high Gender Inequality Index (0.50) shows that women in particular are left out.

**A3-** Because HDI also measures education and life expectancy, not just income, a country that does not invest its oil money in health and schooling scores lower than its GNI alone would predict.

**Common mistake:** “Tessar has a lower HDI because it is a smaller country.” Size has nothing to do with it; the table doesn't even show population. The reason is WHERE the money goes: concentrated oil wealth that isn't spent on schools and health.

**Why the model works:** It names the reason (concentrated wealth), proves it with Tessar's row (85% petroleum, GII 0.50), and connects it to what HDI measures (education and life expectancy).

**D. Explain ONE way a high Gender Inequality Index can slow a country's economic development.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a specific barrier women face: schooling, paid jobs, loans/credit, property, political power, health, early marriage or childbirth
- Explains the effect on the economy: fewer workers or skills, less produced, lower incomes, slower growth

**MODEL ANSWER**

**T1-** A high Gender Inequality Index slows development because it keeps half the population from fully contributing to the economy.
**E2-** In Sulmani (GII 0.58), many girls leave school early to marry, and few women hold paid jobs or can get loans to start businesses.
**A3-** Because educated, working women earn income, have fewer and healthier children, and start businesses, shutting them out shrinks the workforce and the country's skills, so the economy grows more slowly than it could.

**Common mistake:** “Gender inequality is bad for the economy.” That just restates the question. Name a specific thing women are kept from (school, jobs, loans) and explain how that leads to slower growth.

**Why the model works:** It names the general idea (half the population shut out), gives a specific barrier (school, jobs, loans in Sulmani), and explains the chain to slower growth.

**E. Explain ONE economic risk for a country whose exports depend heavily on a single commodity.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a risk: price shock or volatility, running out of the resource, falling demand, or the resource curse
- Explains the effect on the country: lost income or revenue, budget cuts, debt, unemployment, recession

**MODEL ANSWER**

**T1-** Depending on one commodity is risky because the whole economy rises and falls with that product's world price, which the country cannot control.
**E2-** Tessar gets 85% of its export earnings from petroleum, so if oil prices crash, as they did from over $100 a barrel in 2014 to under $30 in 2016, government revenue and jobs collapse at the same time.
**A3-** Because the country has few other exports to fall back on, a price shock can lead to budget cuts, debt, and unemployment, which is one part of what geographers call the resource curse: resource-rich countries often end up less stable and less developed than expected.

**Common mistake:** “Tessar should sell more things.” That's advice, not a risk. The question asks what could go WRONG: name the danger (a price crash) and explain how it hurts the country.

**Why the model works:** It names the risk (price shock), gives proof (Tessar's 85% petroleum and the 2014–2016 oil crash), and traces the chain to budget cuts, debt, and job losses.

**F. Explain how a special economic zone (SEZ) or export processing zone could help Sulmani move toward Rostow's take-off stage, AND identify one potential downside.** EXPLAIN

EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:

- Explains what the SEZ offers or attracts: tax breaks, no tariffs, foreign investment, factories, jobs, exports, technology
- Connects it to take-off or industrial growth: moving workers out of farming, a new leading industry, faster growth
- Identifies a downside: low wages, exploitation, pollution, weak protections, lost tax revenue, companies leaving, dependence on foreign firms

MODEL ANSWER

**T1-** A special economic zone could help Sulmani start to industrialize by attracting foreign factories.
**E2-** Inside the zone, the government offers low taxes, no tariffs on imported parts, and ready-built roads and power, so multinational companies open clothing or electronics plants that hire workers leaving farms, much as the Shenzhen SEZ turned a fishing town into a manufacturing giant.
**A3-** Those factories create a new leading sector, export earnings, and job skills that can push Sulmani from Rostow's traditional stage toward take-off; however, the downside is that wages are often low and labor and environmental rules are weak, and if the tax breaks end the companies may simply move to a cheaper country.

**Common mistake:** "An SEZ would give Sulmani factories and jobs." That's a start, but it stops too early. Connect the factories to take-off (workers leave farms, industry grows fast) AND name one downside.

**Why the model works:** It explains what the zone offers (tax breaks, infrastructure, foreign factories), connects it to take-off (new leading sector pulling workers from farms), and names a clear downside (low wages, companies leaving).

**G. Explain ONE limitation of using the data in the table to compare the quality of life of people in these four countries.** EXPLAIN

**EARNS THE POINT IF THE ANSWER DOES ALL OF THESE:**

- Names a limitation: national averages hide regional/urban-rural/gender/rich-poor gaps, informal economy not counted, only one year, PPP issues, or quality of life factors not measured
- Explains why it misleads, ideally with an example from the table

**MODEL ANSWER**

**T1-** One limitation is that every number in the table is a national average for a single year, which hides differences between groups inside each country.
**E2-** For example, Tessar's $31,400 GNI per capita is an average: a small oil elite in the capital may be extremely rich while rural families and women (GII 0.50) live on far less, and Kovar's 55% service workers may be concentrated in a few big cities.
**A3-** Because the averages flatten the gaps between rich and poor regions, genders, and the formal and informal economy, two countries with similar numbers could offer very different lives, so you would need data at a smaller scale to compare quality of life fairly.

**Common mistake:** “The table only has four countries.” That's about how many rows there are, not about what the numbers can tell you. Explain what the averages hide, like gaps between rich and poor regions or men and women.

**Why the model works:** It names the limitation (national averages for one year), shows what they hide with a specific example (Tessar's oil elite vs. rural families), and explains why that makes comparing quality of life unreliable.

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